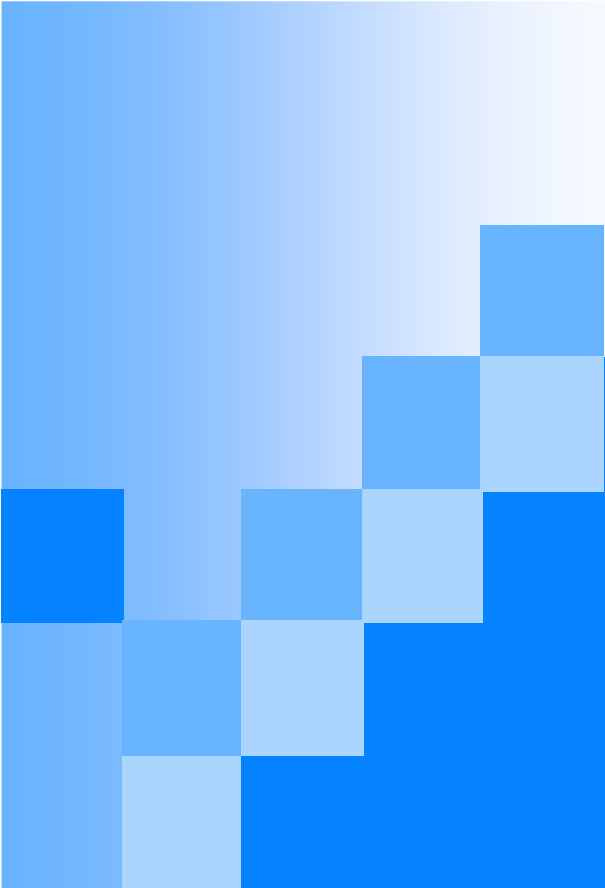




Java EE

Web Layer

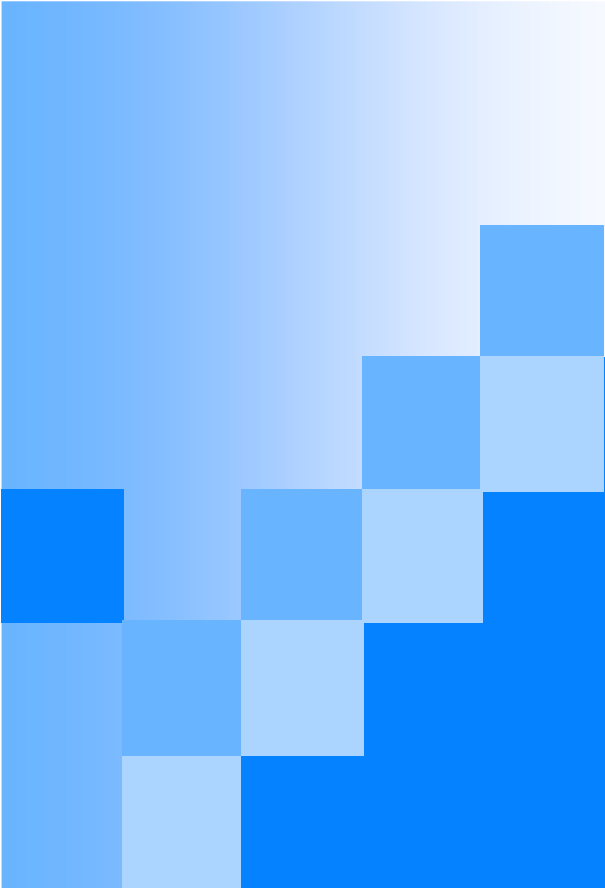
Olivier Planson

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Plan

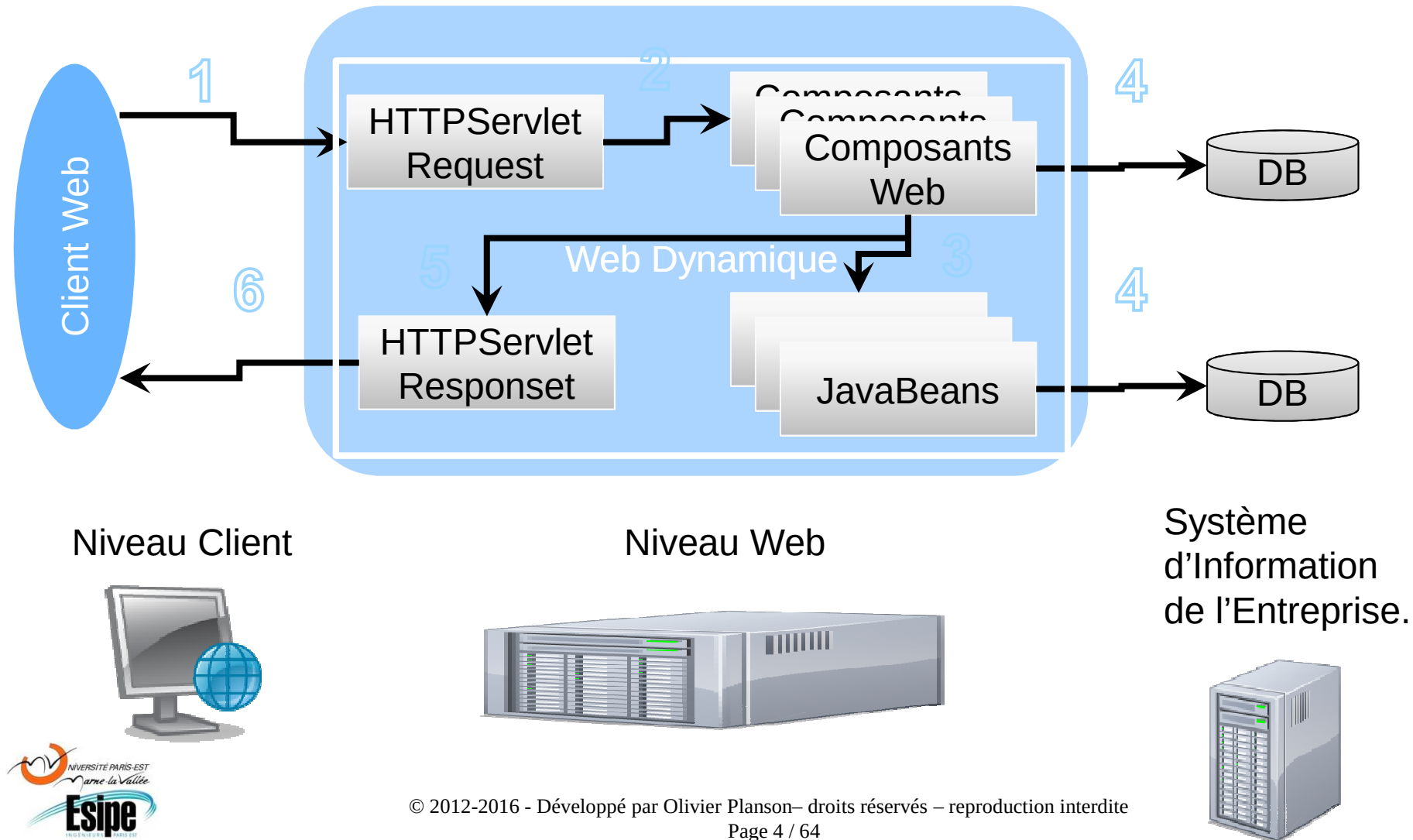
- Introduction
- Le JDBC
- Les Servlets
- Les JSP
- Configuration d'un serveur d'application Web
- Les JSF

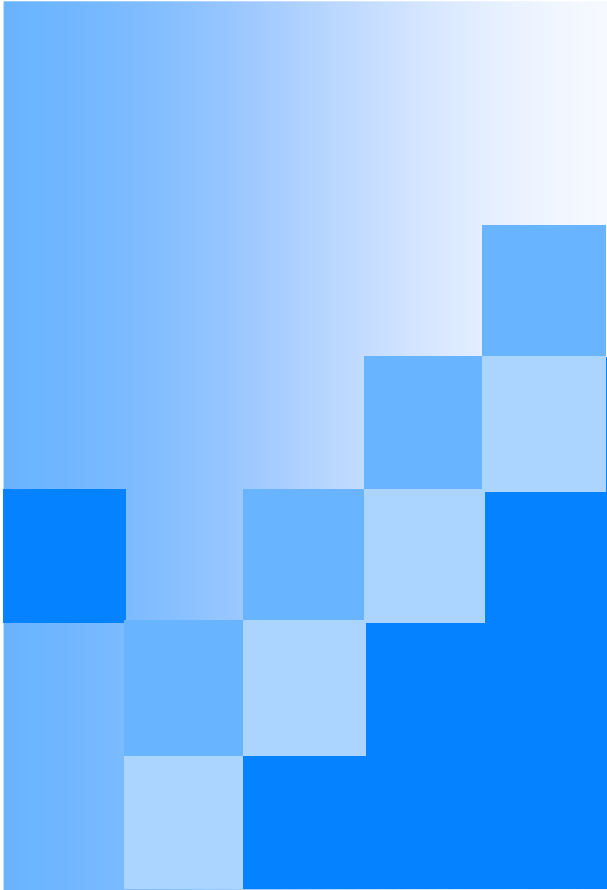


Introduction

"Web Application Architecture"

Introduction

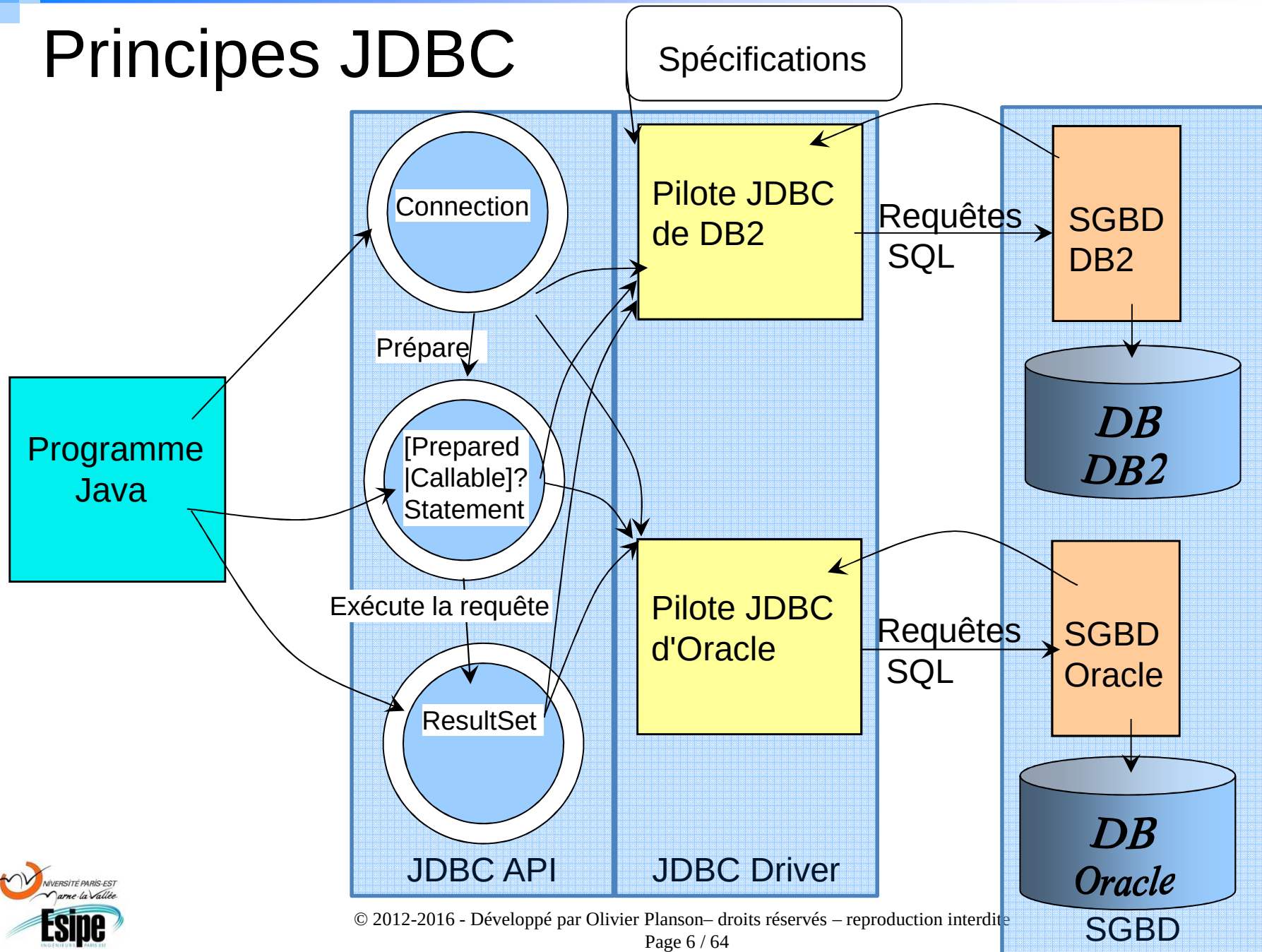




Le Java DataBase Connectivity

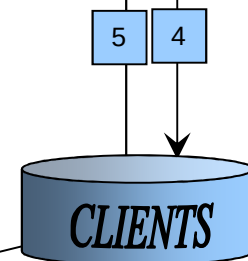
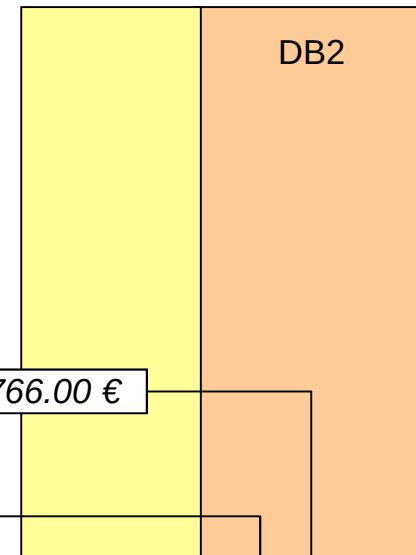
JDBC

Principes JDBC



Exemple JDBC

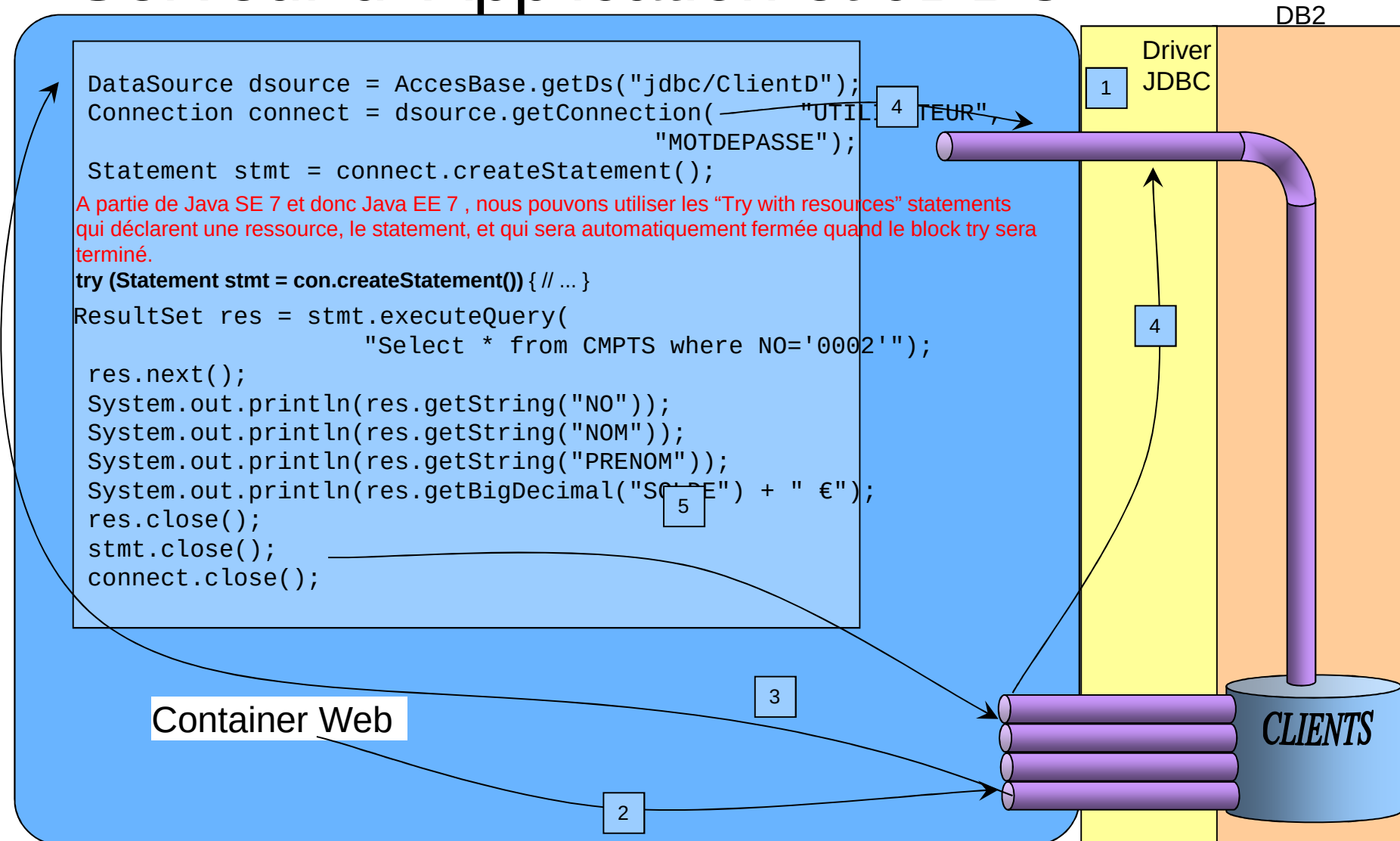
```
Class.forName("COM.ibm.db2.jdbc.app.DB2Driver");  
Connection connect = DriverManager.getConnection(  
    (Coûteux) "jdbc:db2:CLIENTS",  
    "UTILISATEUR",  
    "MOTDEPASSE");  
Statement statmt = connect.createStatement();  
ResultSet res = statmt.executeQuery(  
    "Select * from CMPTS where NO='0002'");  
res.next();  
System.out.println(res.getString("NO"));  
System.out.println(res.getString("NOM"));  
System.out.println(res.getString("PRENOM"));  
System.out.println(res.getBigDecimal("SOLDE") + " €");  
res.close();  
statmt.close();  
connect.close();
```

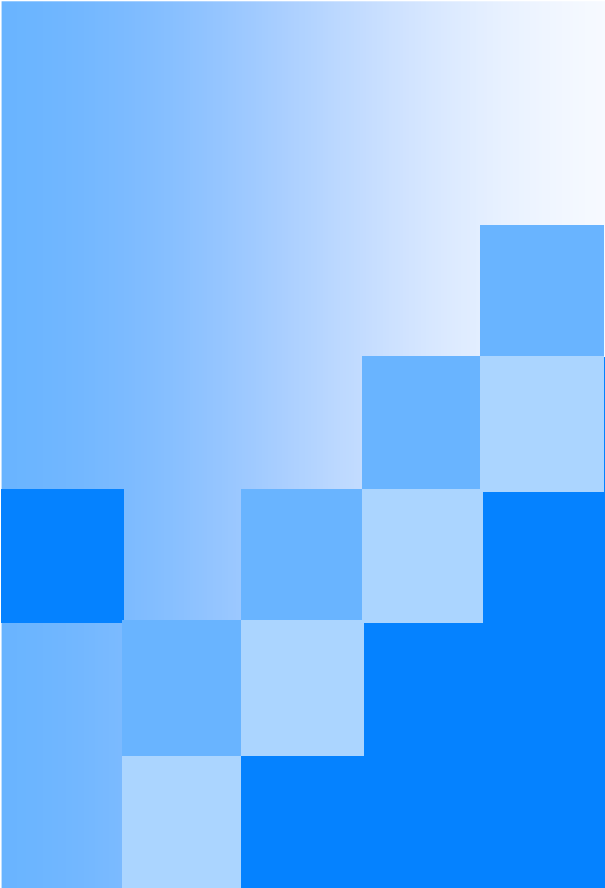


0002
Magnes
Charles
3766.00 €

CMPTS			
NO	NOM	PRENOM	SOLDE
0001	Alban	Loic	3360.57
0002	Magnes	Charles	3766.00
0003	Lemaire	Julie	3000.00
0004	Ronchon	Jacques	6890.90

Serveur d'Application et JDBC





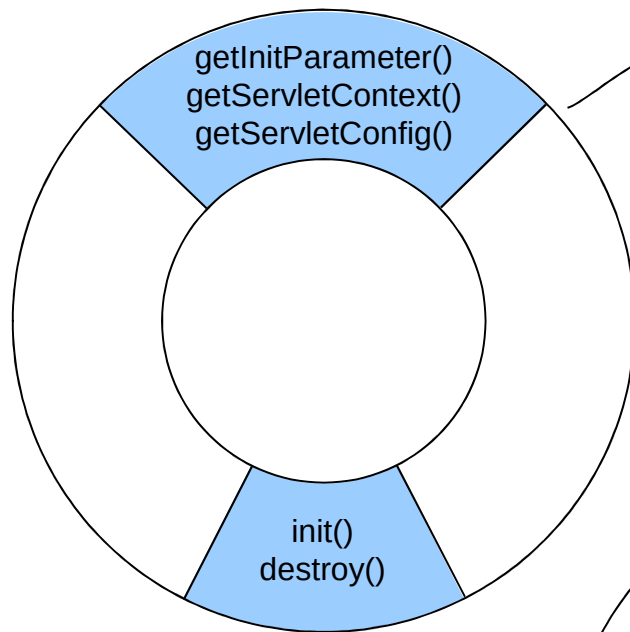
Les Servlets

"JST"



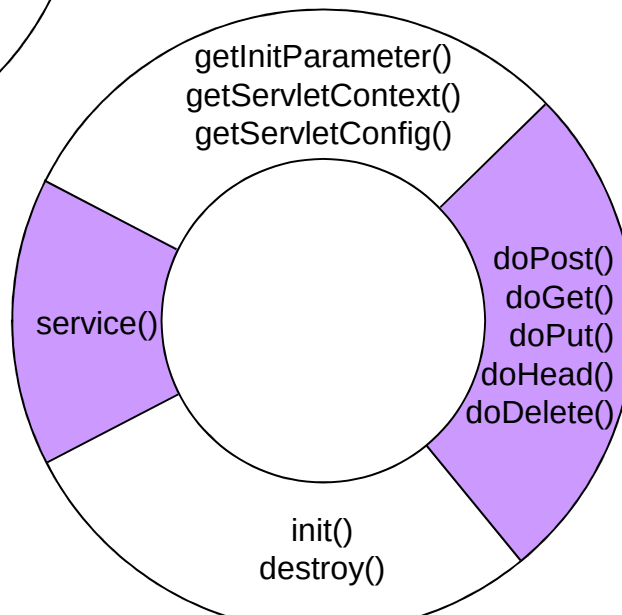
La Construction d'une Servlet

GenericServlet

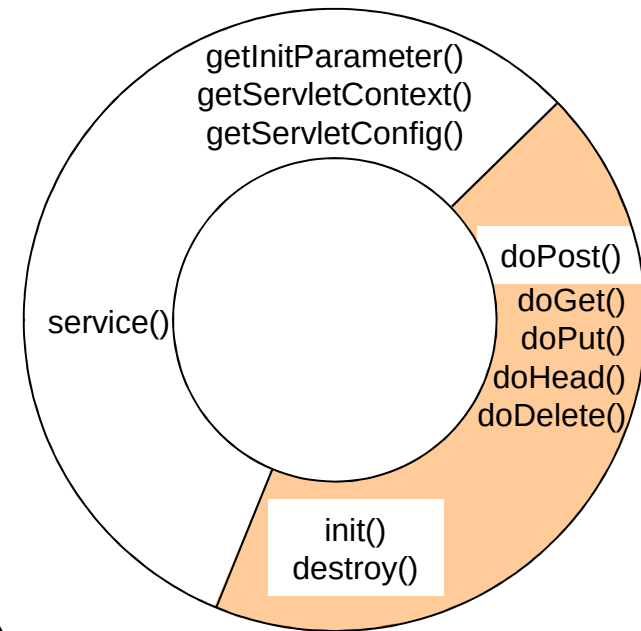


héritage

HttpServlet

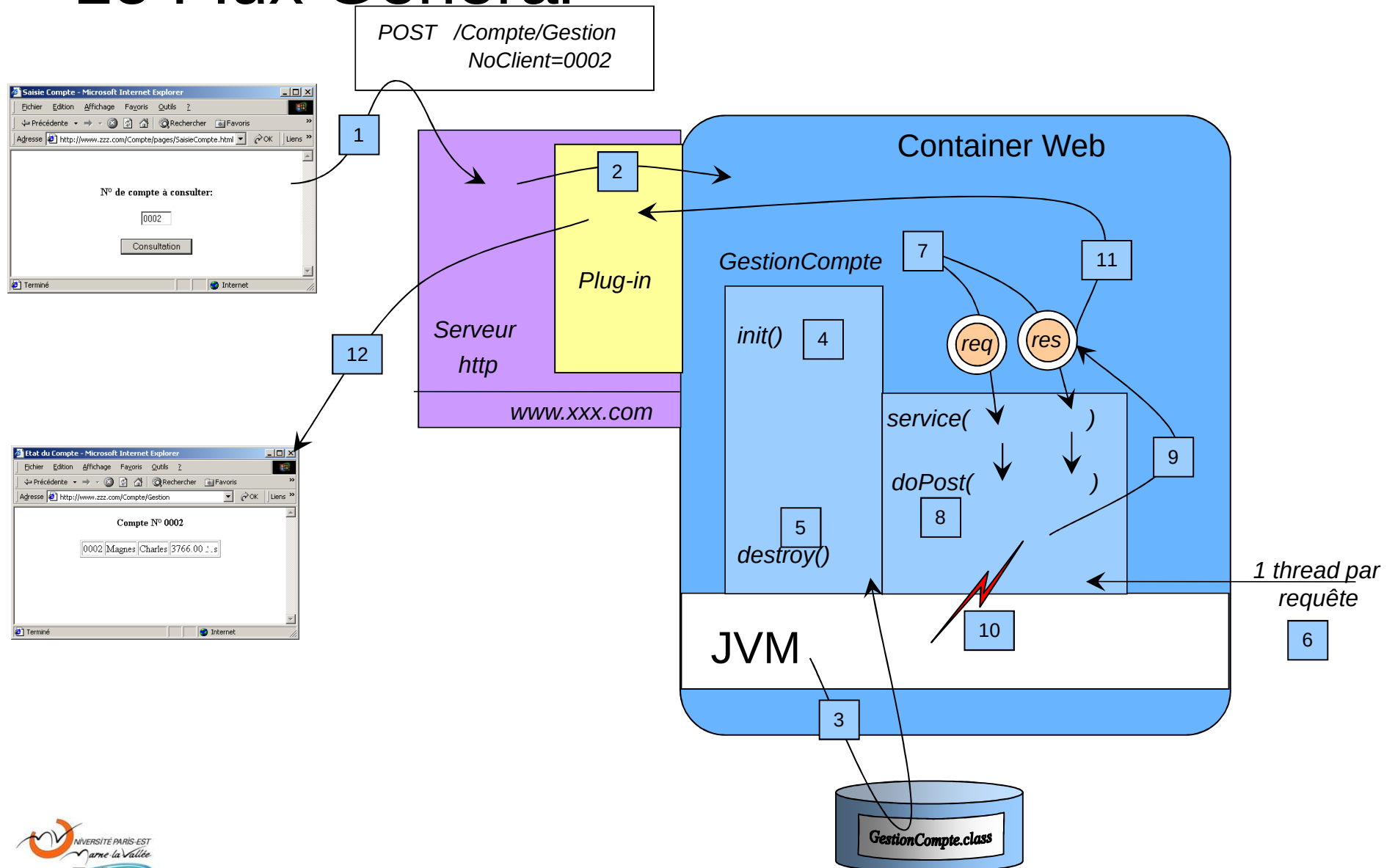


MaServlet



héritage

Le Flux Général



Flux Détaillé (saisie du client)

Saisie Compte - Microsoft Internet Explorer

Fichier Edition Affichage Favoris Outils ?

Précédente → Recherche Favoris

Adresse http://www.zzz.com/Compte/pages/SaisieCompte.html OK

N° de compte à consulter:

Consultation

Terminé Internet

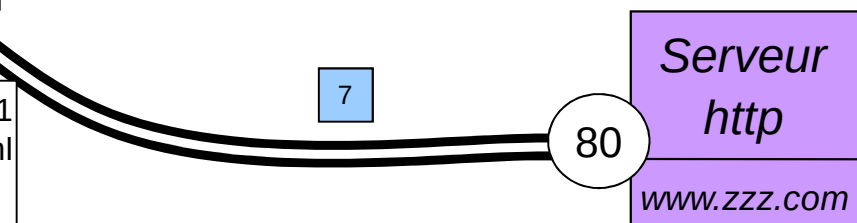
Navigateur

```
<HTML>
<HEAD>
<TITLE> Saisie Compte </TITLE>
</HEAD>
<BODY>
<CENTER>
<FORM action="/Compte/Gestion" method="Post"><BR>
<BR><H4>N° de compte à consulter:</H4>
<INPUT size="4" type="text" maxlength="4"
name="NoDeCompte">
<BR><BR>
<CENTER><INPUT type="submit" name="Demande"
value="Consultation">
</CENTER>
</FORM>
</CENTER>
</BODY>
</HTML>
```

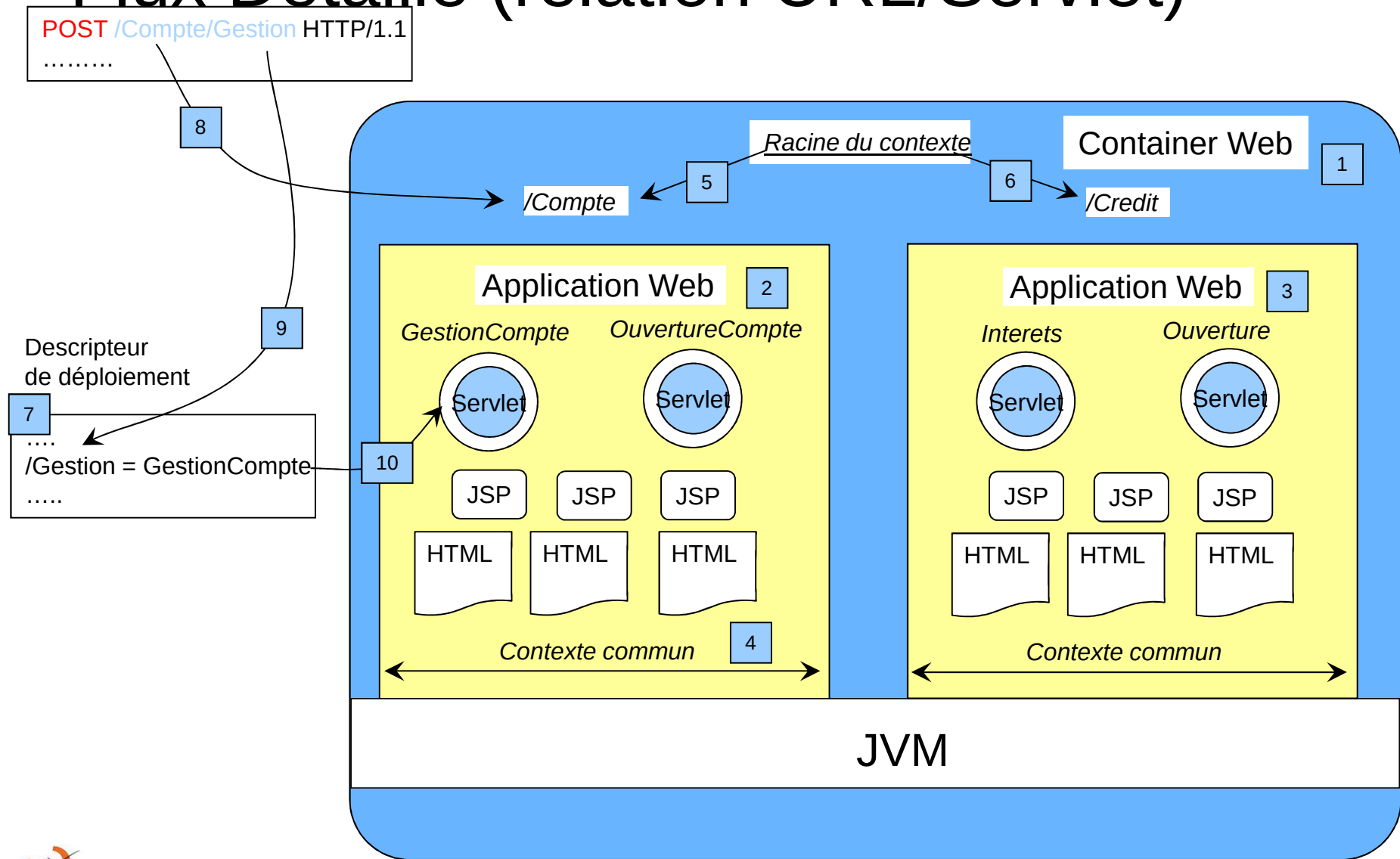
POST /Compte/Gestion HTTP/1.1
referer: http://www.zzz.com/Compte/pages/SaisieCompte.html
Host : www.zzz.com
.....
NoDeCompte=0002

OU

GET /Compte/Gestion?NoDeCommpte=0002&...&... HTTP/1.1
referer: http://www.zzz.com/Compte/pages/SaisieCompte.html
Host : www.zzz.com
.....



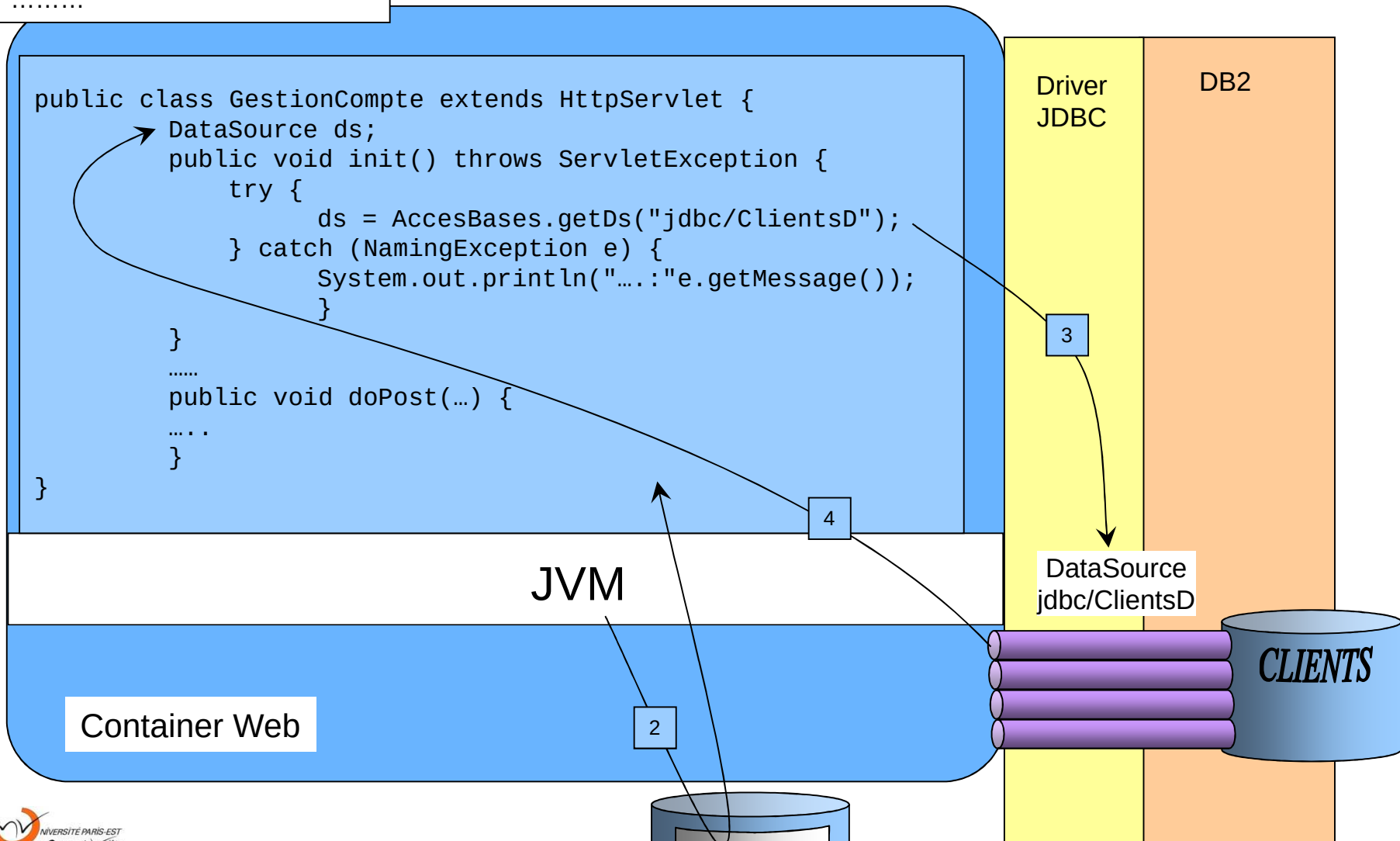
Flux Détaillé (relation URL/Servlet)



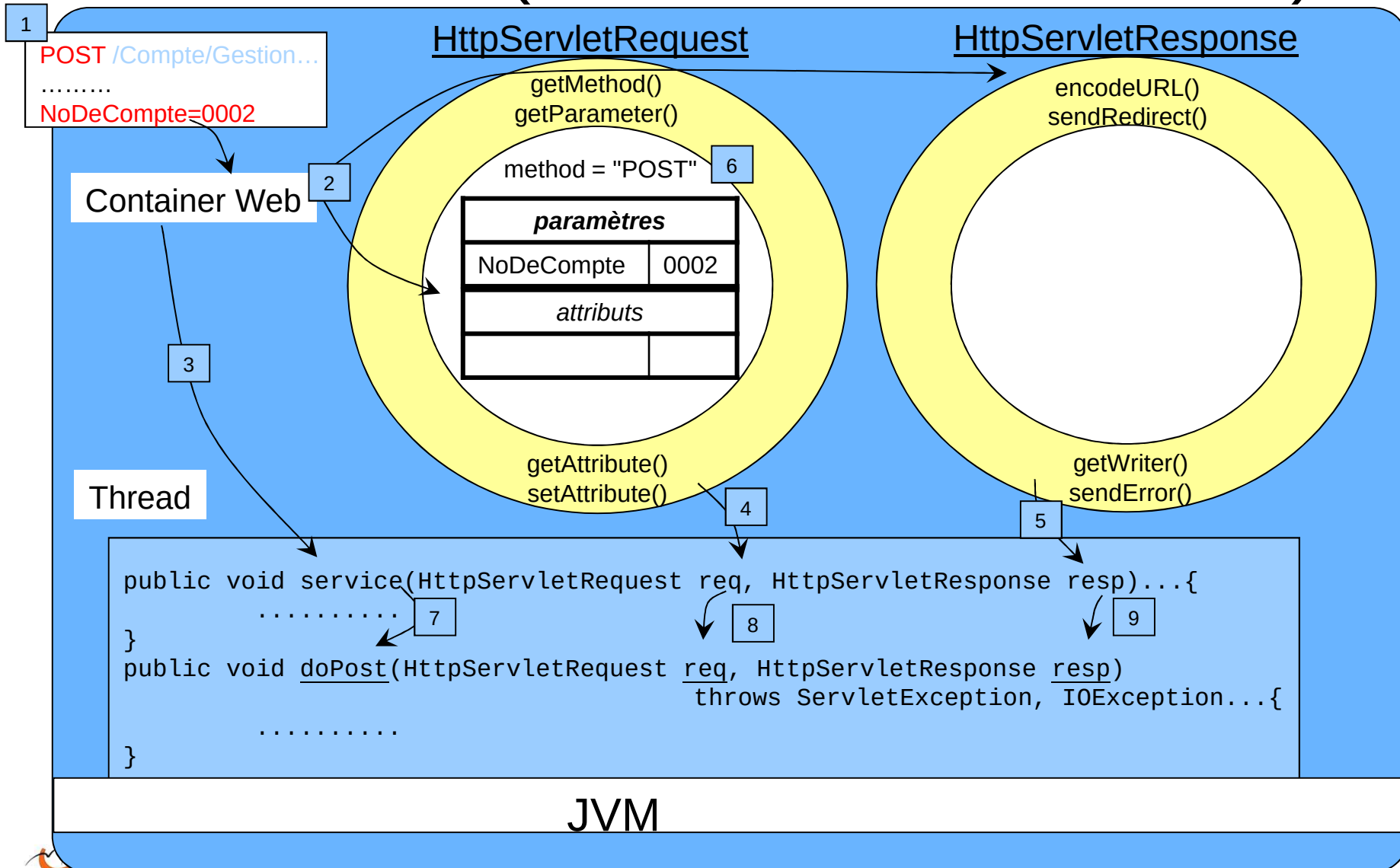
Flux Détaillé - instantiation Servlet

1

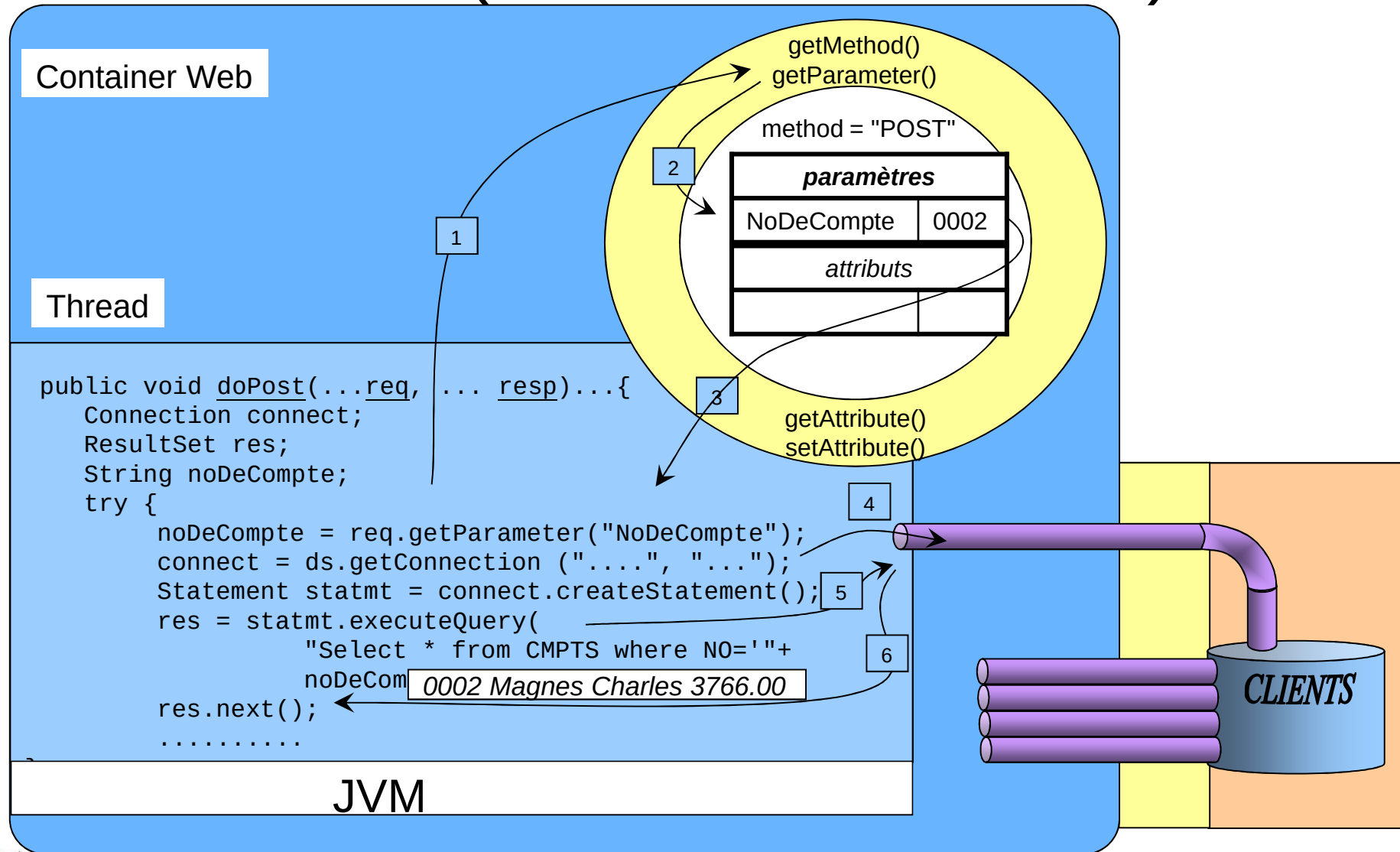
POST /Compte/Gestion HTTP/1.1



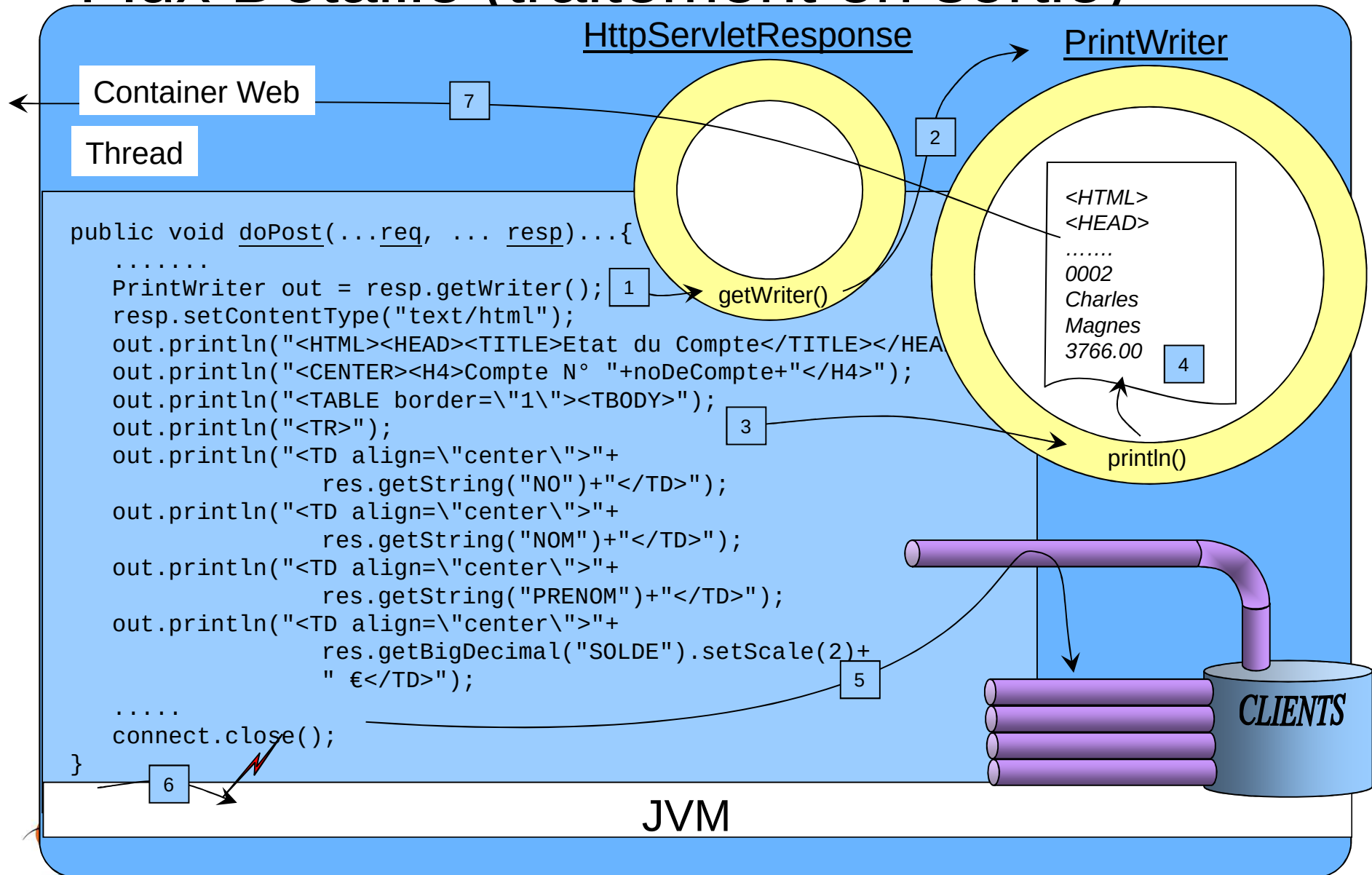
Flux Détaillé (instanciation du Thread)



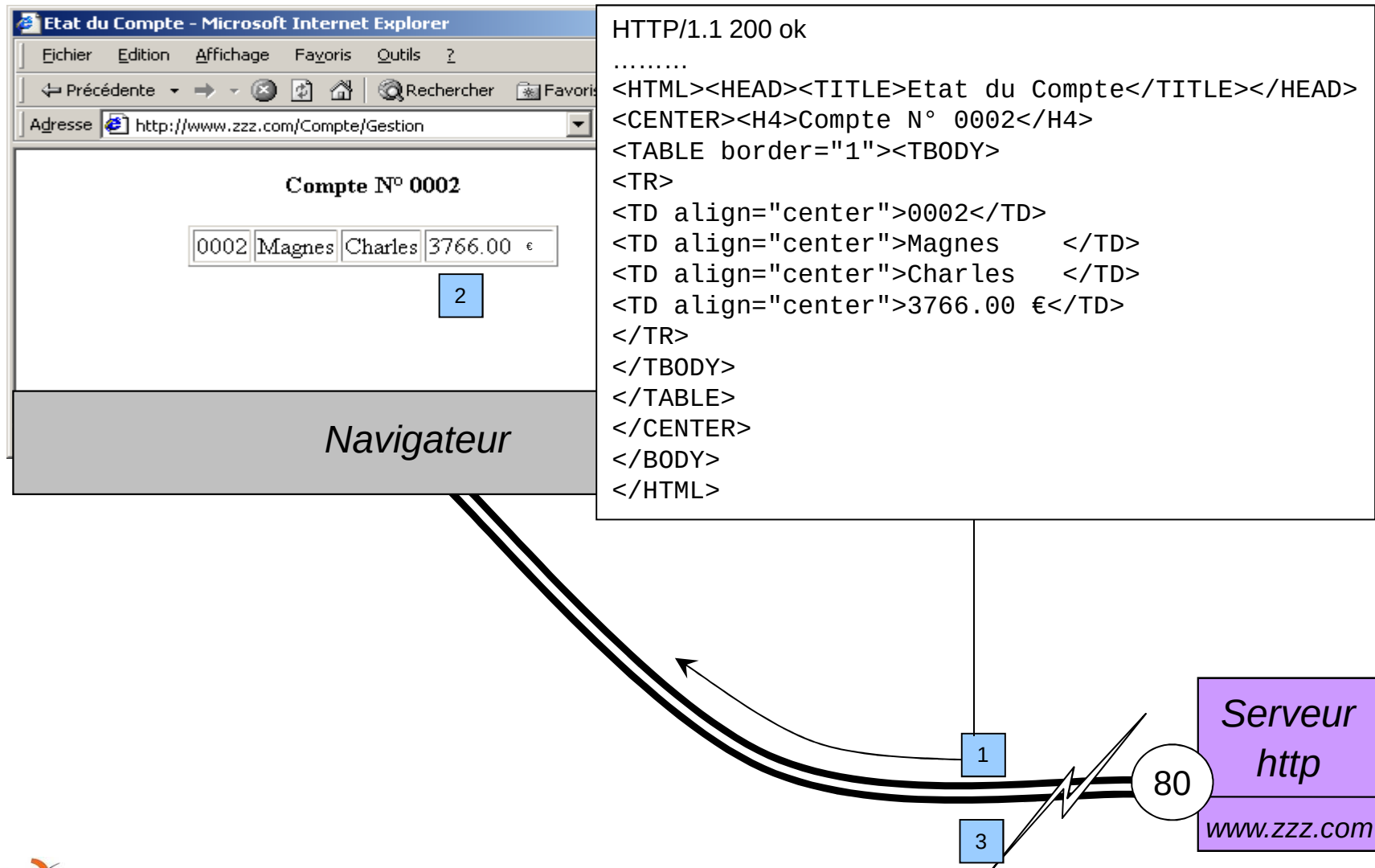
Flux Détaillé (traitement en entrée)



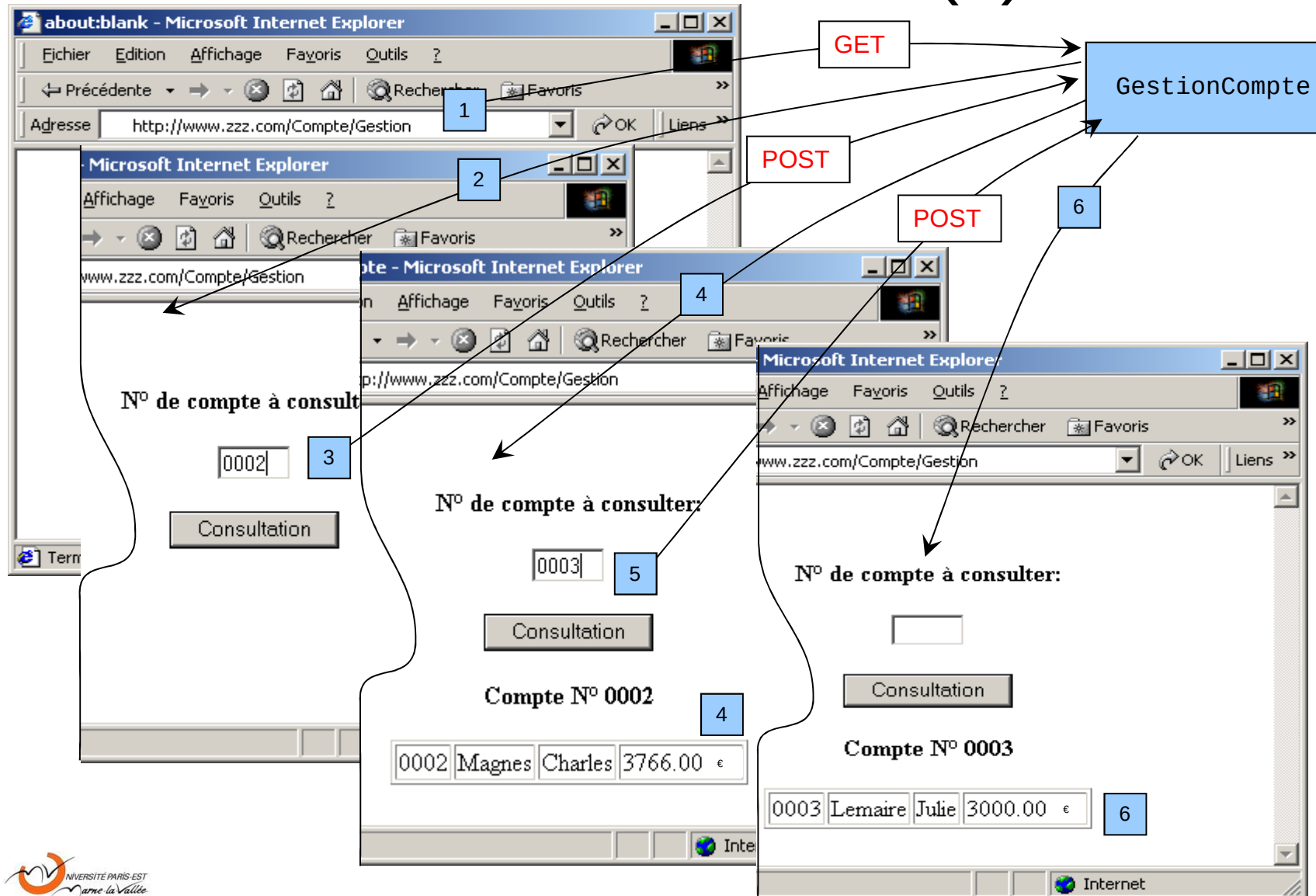
Flux Détaillé (traitement en sortie)



Flux Détaillé (retour du résultat)



Amélioration de la Servlet (1)



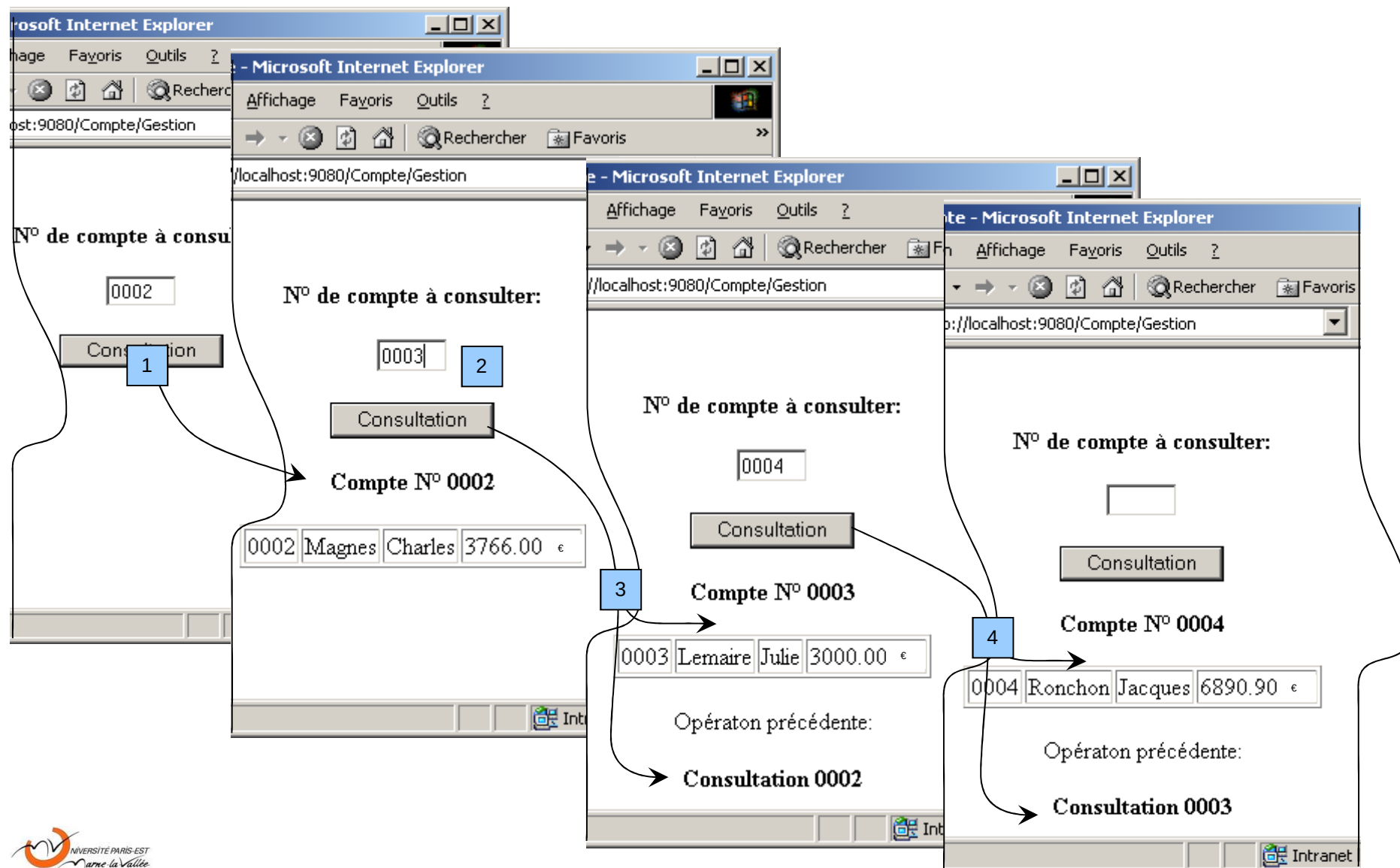
Amélioration de la Servlet (2)

```
public void doGet(HttpServletRequest req, HttpServletResponse resp)
    throws ServletException, IOException {
    doPost(req, resp);
}

public void doPost(...req, ... resp)...{
    .....
    out.println("<HTML><HEAD><TITLE>Etat du Compte</TITLE></HEAD>");
    out.println("<CENTER>");
    out.println("<FORM action=\""/Compte/Gestion\" method=\"Post\"><BR>");
    out.println("<BR><H4>N° de compte à consulter:</H4><INPUT size=\"4\" type=\"text\"
        maxlength=\"4\" name=\"NoDeCompte\"><BR><BR>");
    out.println("<CENTER><INPUT type=\"submit\" name=\"Demande\"
        value=\"Consultation\"></CENTER>");
    out.println("</FORM>");
    if (req.getMethod().equals("POST")) {
        ..... Affichage de la table de résultats
    }
    out.println("</CENTER>");
    out.println("</BODY>");
    out.println("</HTML>");
    connect.close();
    .....
}
```

Diagram annotations: A box labeled '1' is placed over the `doPost(req, resp);` line in the `doGet` method. An arrow points from this box to the `doPost` method signature. Another box labeled '2' is placed over the `Affichage de la table de résultats` line in the `doPost` method.

Besoin de Session

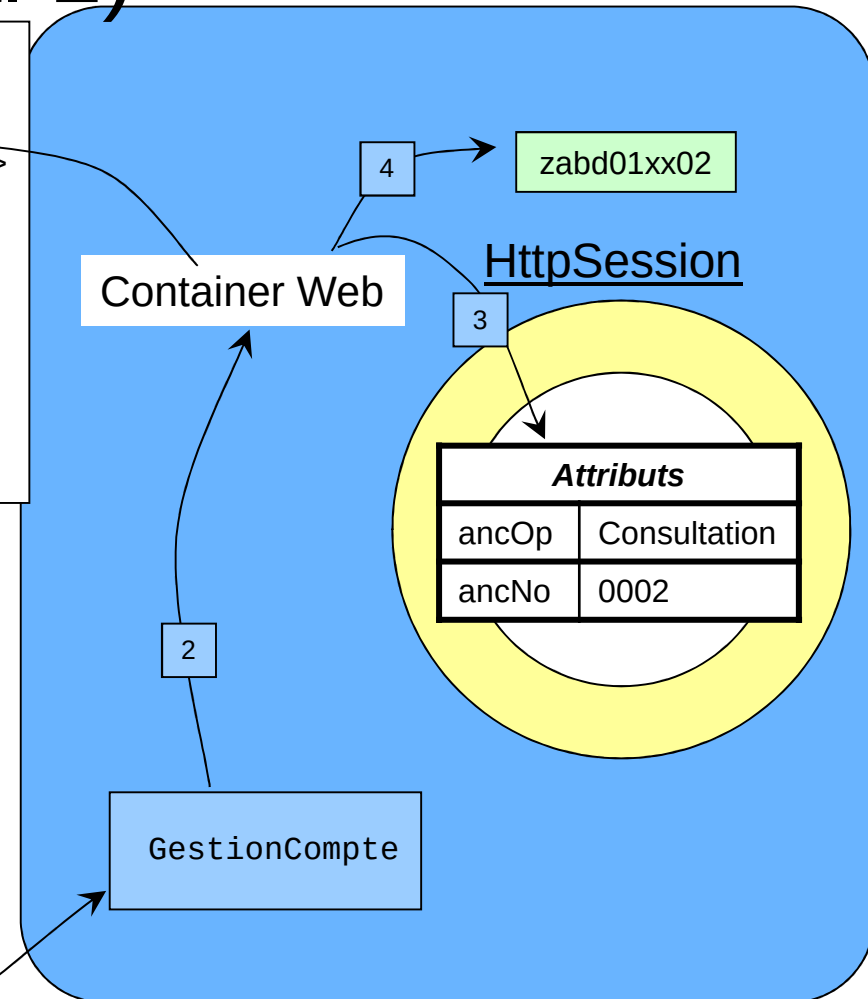


Session (flux général 1)

```
HTTP/1.1 200 ok
set-Cookie2: jsessionid="zabd01xx02"
```

```
.....
<HTML><HEAD><TITLE>Etat du Compte</TITLE></HEAD>
<CENTER><H4>Compte N° 0002</H4>
<TABLE border="1"><TBODY>
<TR>
<TD align="center">0002</TD>
<TD align="center">Magnes    </TD>
<TD align="center">Charles   </TD>
<TD align="center">3766.00 €</TD>
.....
```

```
POST /Compte/Gestion HTTP/1.1
referer: http://www.zzz.com/Compte/pages/SaisieCompte.html
Host : www.zzz.com
.....
NoDeCompte=0002
```



Session (flux général 2)

Microsoft Internet Explorer

Affichage Favoris Outils ?

Rechercher Favoris

localhost:9080/Compte/Gestion

N° de compte à consulter:

Consultation

Compte N° 0002

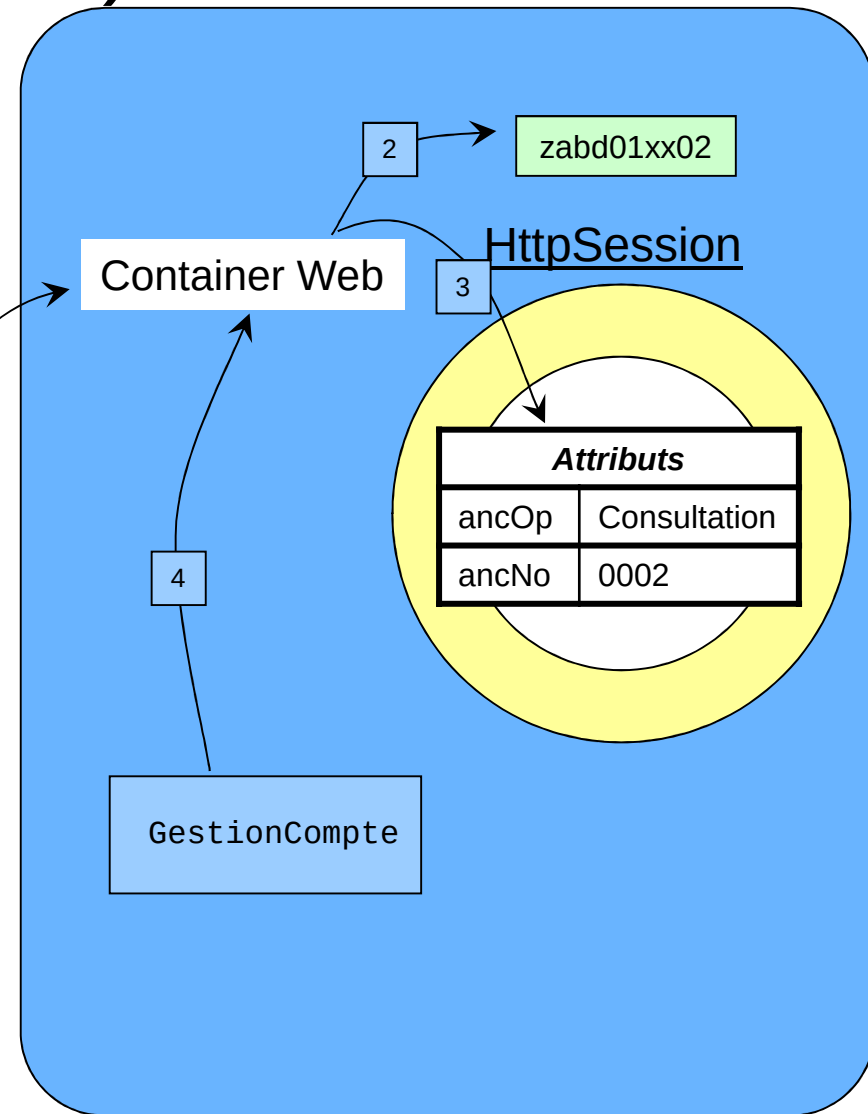
Navigateur

1

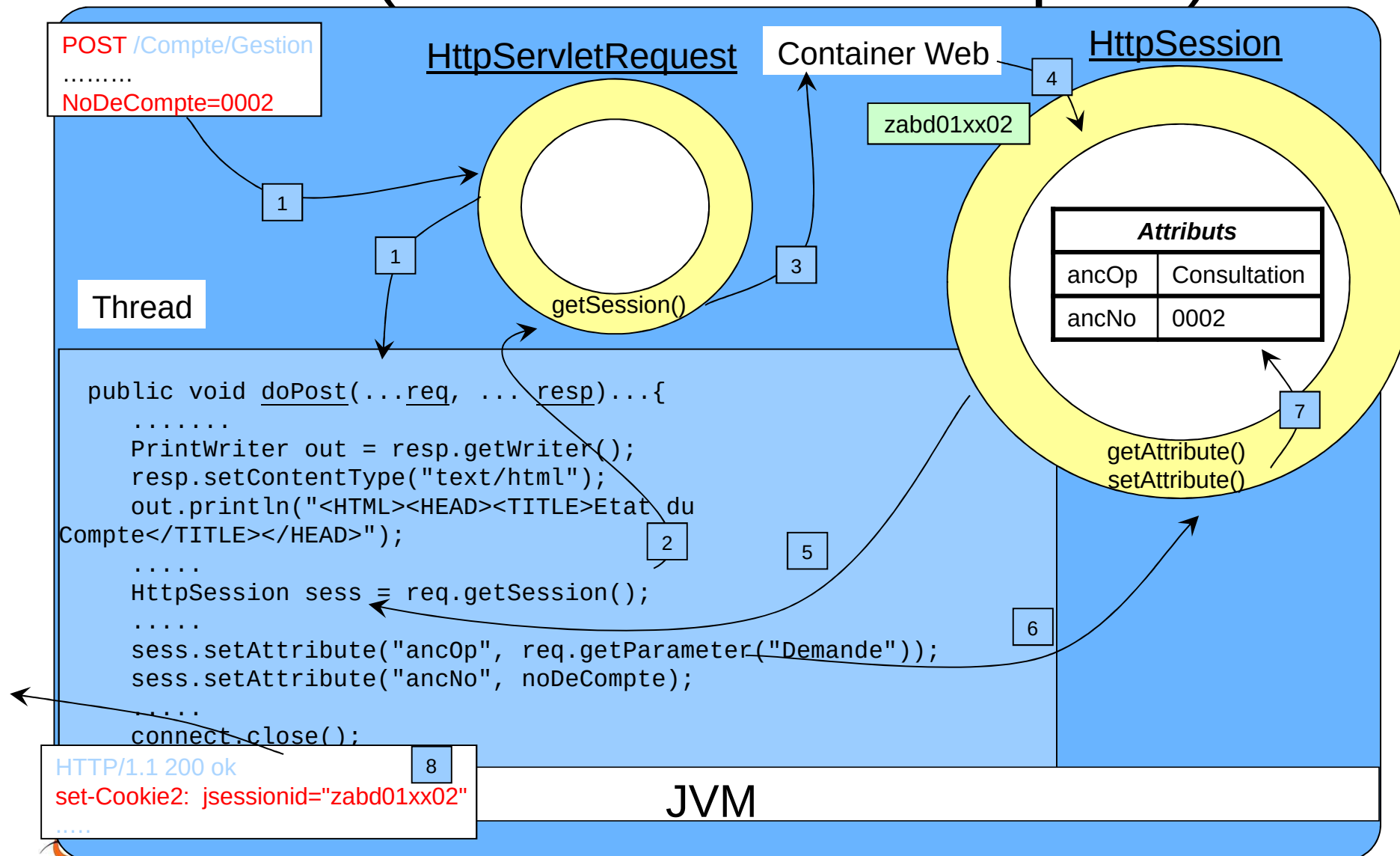
POST /Compte/Gestion HTTP/1.1
 Cookie: jsessionid="zabd01xx02"
 referer: http://www.zzz.com/Compte/pages/SaisieCompte.html
 Host : www.zzz.com

.....

NoDeCompte=0003



Session (flux détaillé-1ère requête)



Session (flux détaillé-après la réponse)

Microsoft Internet Explorer

Affichage Favoris Outils ?

Rechercher Favoris

/localhost:9080/Compte/Gestion

N° de compte à consulter:

0003

Consultation

Compte N° 0002

0002 Magnes Charles 3766.00 €

Navigateur

Cookie: jsessionid="zabd01xx02"

1



2

zabd01xx02

Attributs	
ancOp	Consultation
ancNo	0002

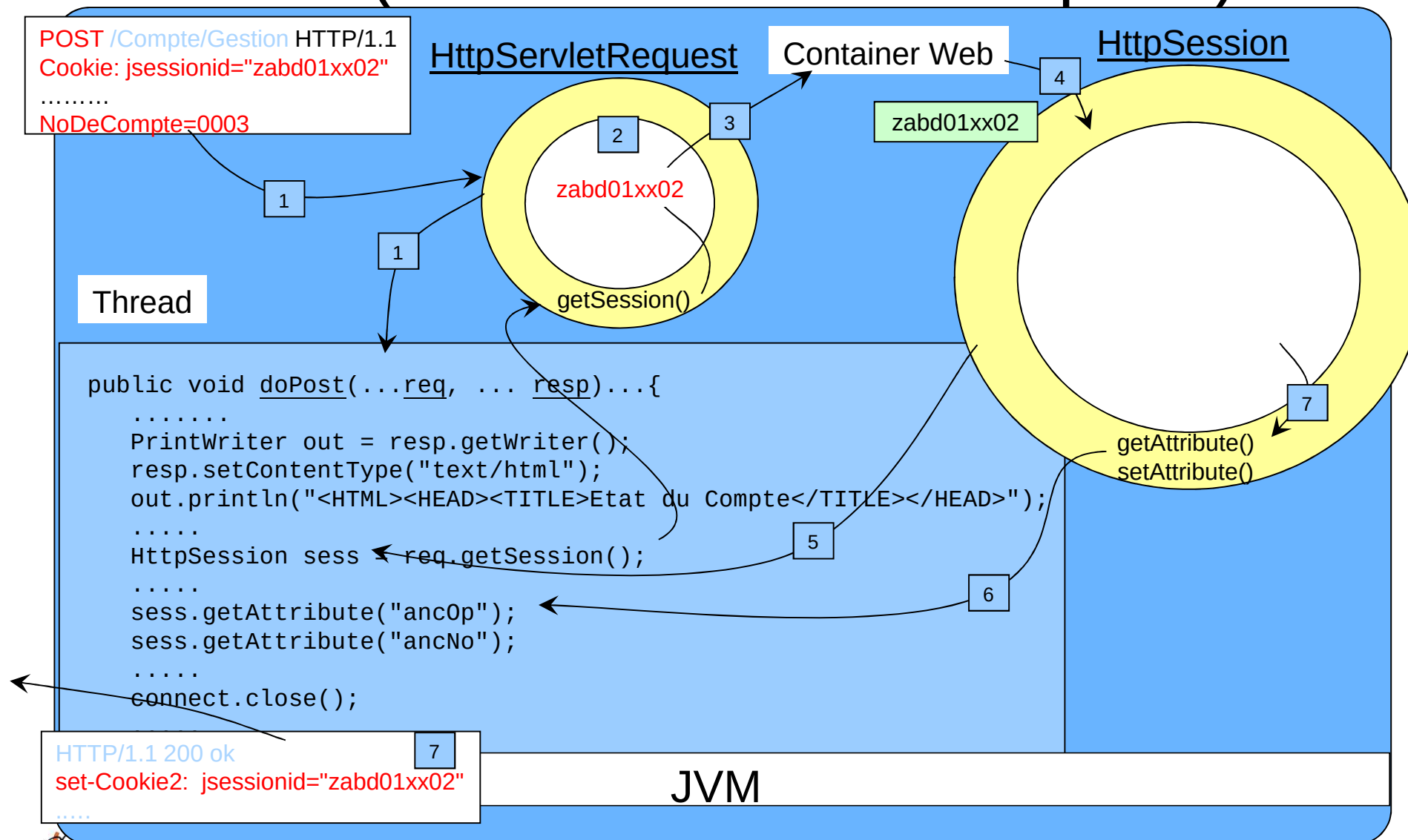
getAttribute()
setAttribute()

HttpSession

Container Web

JVM

Session (flux détaillé-2ème requête)



Session (la logique complète)

HttpServletRequest

zabd01xx02

getSession()

```
public void doPost(...req, ... resp)...{
    .....
    PrintWriter out = resp.getWriter();
    resp.setContentType("text/html");
    out.println("<HTML><HEAD><TITLE>Etat du Compte</TITLE></HEAD>");
    .....
    HttpSession sess = req.getSession();
    String ancOp = (String) sess.getAttribute("ancOp");
    String ancNo = (String) sess.getAttribute("ancNo");
    if (ancOp != null && ancNo != null){
        out.println("<BR>Opératon précéd e:<BR>");
        out.println("<H4>"+ancOp+" "+ancNo+"</H4>");
    }
    sess.setAttribute("ancOp", req.getParameter("Demande"));
    sess.setAttribute("ancNo", noDeCompte);
    .....
    connect.close();
    .....
}
```

Attributs	
ancOp	Consultation
ancNo	0002

getAttribute()
setAttribute()

HttpSession

Session (destruction)

Container Web



Session Time out

1

zabd01xx02

HttpSession

setMaxInactiveInterval()

Attributs

ancOp	Consultation
ancNo	0002

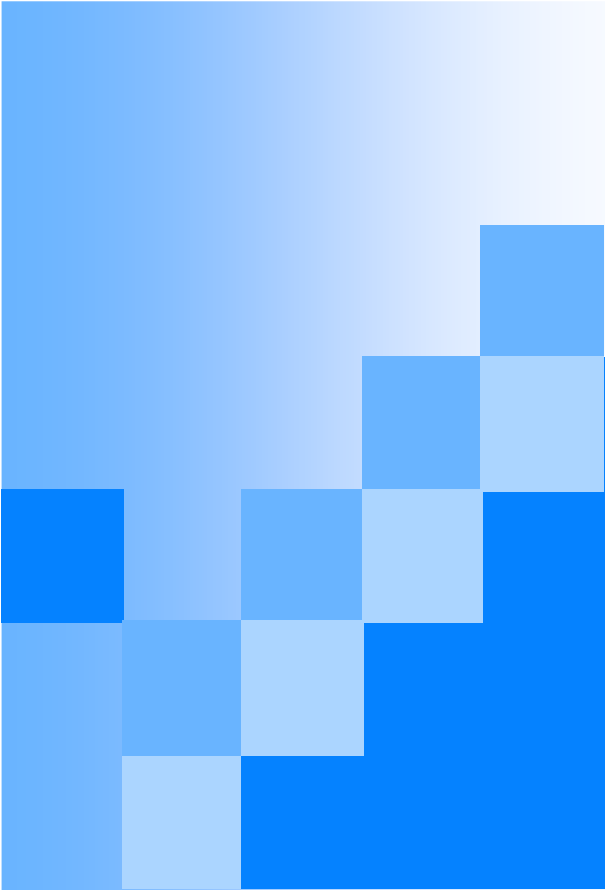
invalidate()

```
public void doPost(...req, ... resp)...{  
    .....  
    sess.setMaxInactiveInterval(40);  
    .....  
    sess.invalidate();  
    .....  
}
```

2

3

JVM



Les JavaBeans



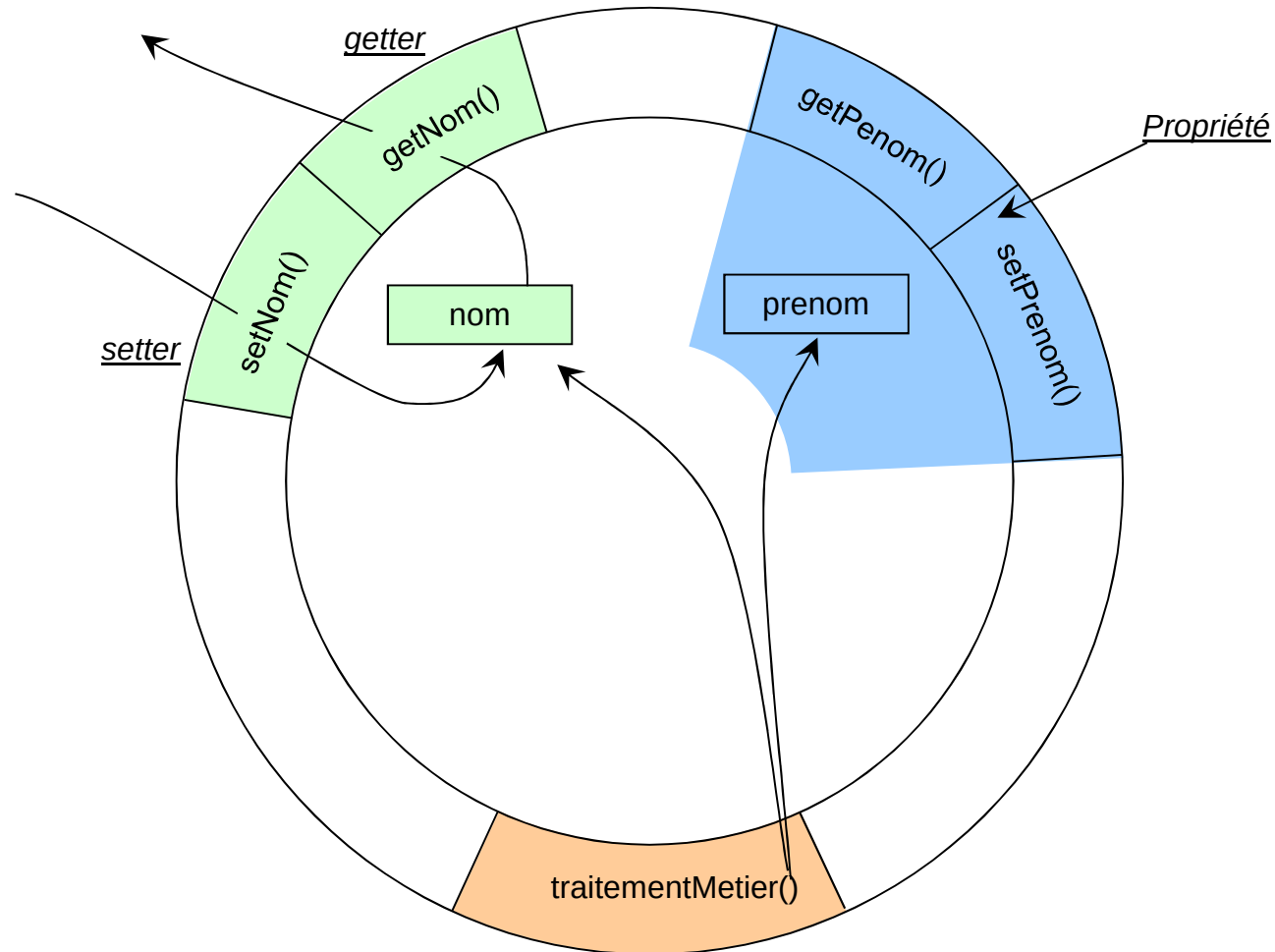
Pourquoi ?

- ❑ **Isoler le traitement métier dans un objet Indépendant.**



Maintenance plus souple

Qu'est-ce qu'un JavaBean



Le JavaBean ConsultationDeCompte

```
public class ConsultationDeCompte {  
    private DataSource ds;  
    private String noDeCompte, nom, prenom, solde;  
  
    public String getNom() {  
        return nom;  
    }  
    public String getPrenom() {  
        return prenom;  
    }  
    public String getSolde() {  
        return solde;  
    }  
    public void setDs(DataSource ds) {  
        this.ds = ds;  
    }  
    public void setNoDeCompte(String noDeCompte) {  
        this.noDeCompte = noDeCompte;  
    }  
    public void consultation() throws SQLException {  
        Connection connect;  
        ResultSet res;  
        connect = ds.getConnection("UTILISATEUR", "MOTDEPASSE");  
        Statement statmt = connect.createStatement();  
        res = statmt.executeQuery(  
            "Select * from CMPTS where NO='" + noDeCompte + "'");  
        res.next();  
        nom = res.getString("NOM");  
        prenom = res.getString("PRENOM");  
        solde = res.getBigDecimal("SOLDE").setScale(2).toString();  
    }  
}
```

Déclaration des variables

Getters et Setters

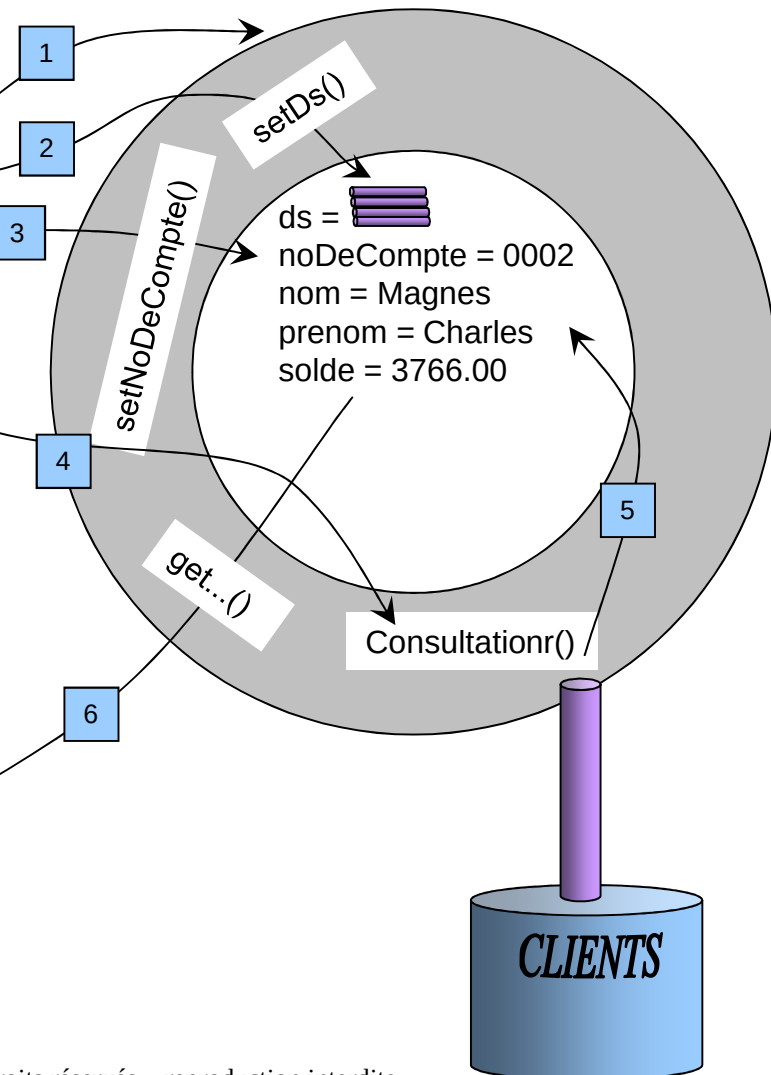
Accès JDBC

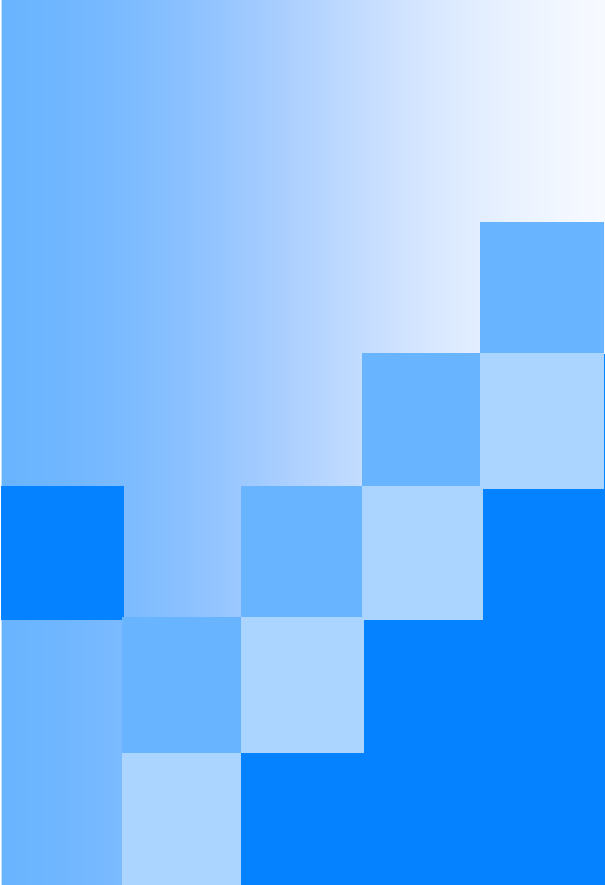
Alimentation des résultats

Modification de la Servlet

```
public void doPost(...req, ... resp){
    String noDeCompte = null;
    ConsultationDeCompte consult = null;
    try {
        noDeCompte = req.getParameter("NoDeCompte");
        if (noDeCompte != null) {
            consult = new ConsultationDeCompte();
            consult.setDs(ds);
            consult.setNoDeCompte(noDeCompte);
            consult.consultation();
        }
        PrintWriter out = resp.getWriter();
        resp.setContentType("text/html");
        out.println("<HTML><HEAD><..>");
        .....
        out.println("<TD align=\"center\">"
            + noDeCompte
            + "</TD>");
        out.println("<TD align=\"center\">"
            + consult.getNom()
            + "</TD>");
        out.println("<TD align=\"center\">"
            + consult.getPrenom()
            + "</TD>");
        out.println("<TD align=\"center\">"
            + consult.getSolde()
            + " Frs</TD>");
        .....
    }
}
```

ConsultationDeCompte





Les Java Server Pages



Inconvénients de la Servlet

- ❑ **Impossibilité d'utiliser un éditeur HTML**
 - **Nécessité d'une connaissance approfondie d'HTML.**
 - **C'est le Programmeur Java qui doit composer la page.**
- ❑ **Toute modification de la page implique la modification de la Servlet**
 - **Maintenance complexe:**
 - **Modification du code source + compilation + instantiation.**

★ **Solution** ➡ **Le JSP**



Structure d'une JSP

MaJSP.jsp

```
<HTML>
<HEAD>
.....
</HEAD>
<BODY>
<H1>....</H1>
```

```
.....
<%= expression java %>
```

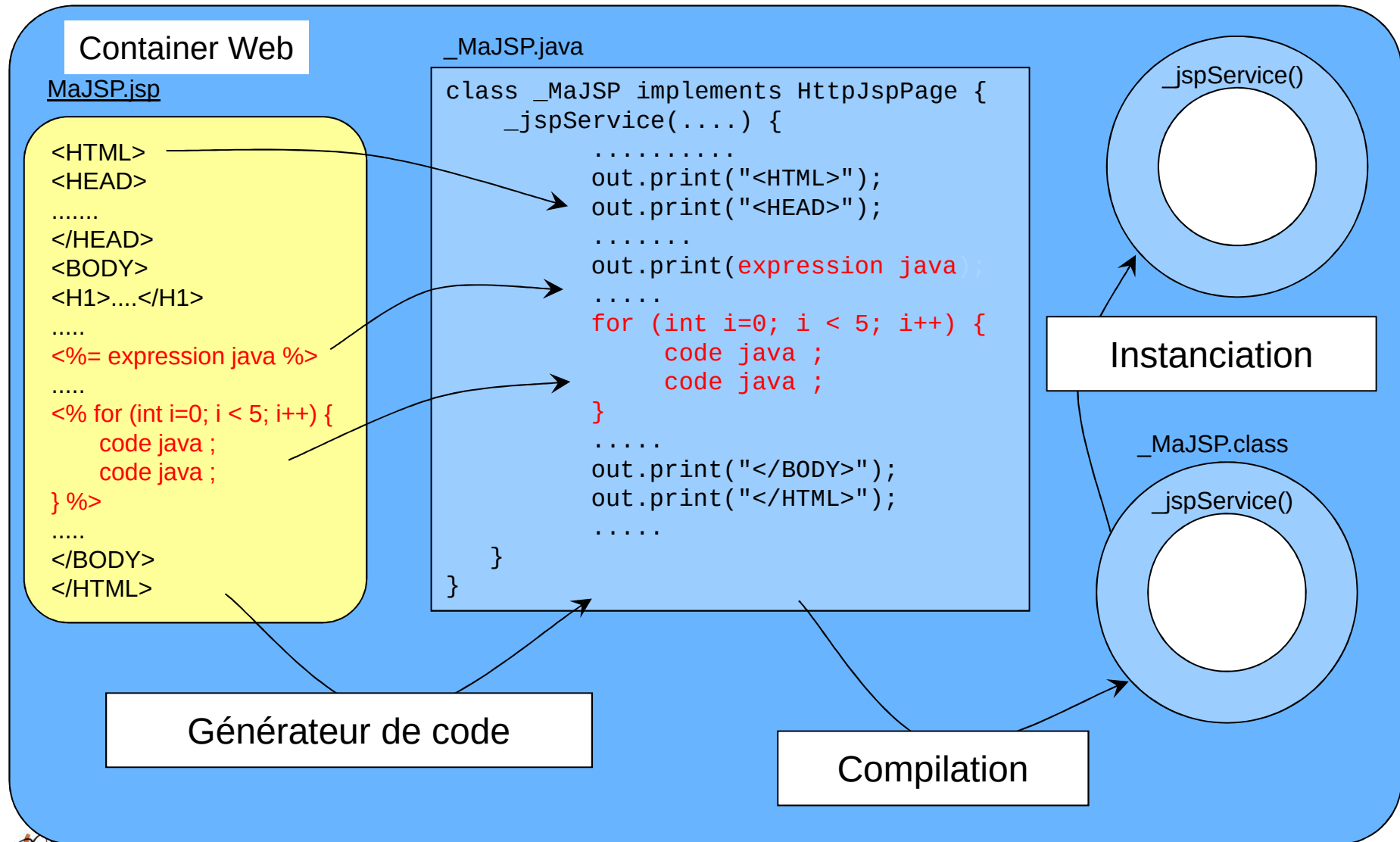
Expression java = code Java renvoyant un résultat venant se substituer à la balise

```
.....
<% for (int i=0; i < 5; i++) {
    code java ;
    code java ;
} %>
```

Scriptlet = code Java ne renvoyant pas obligatoirement un résultat

```
.....
</BODY>
</HTML>
```

L'Execution d'une JSP



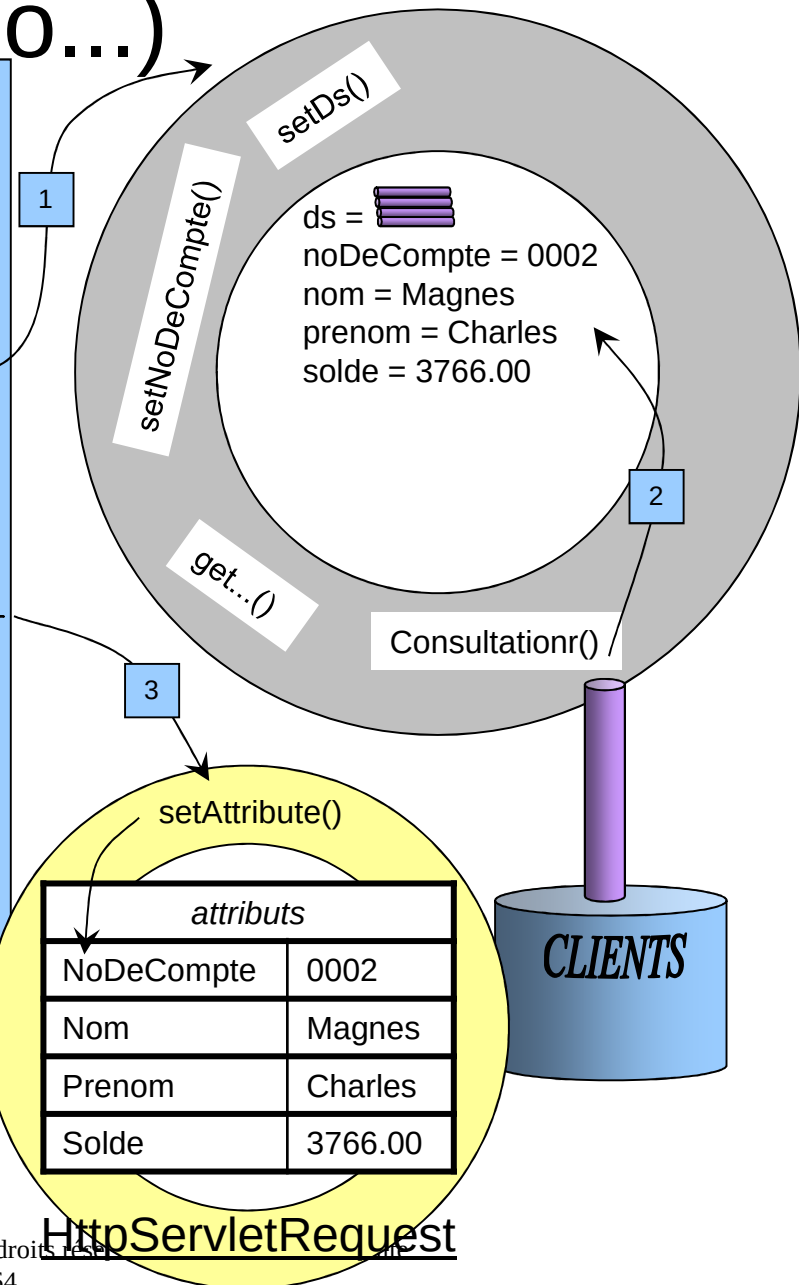
La Servlet (1er scénario...)

```

public void doPost(...req, ... resp)...{
    String noDeCompte = null;
    ConsultationDeCompte consult = null;
    try {
        noDeCompte = req.getParameter("NoDeCompte");
        if (noDeCompte != null) {
            consult = new ConsultationDeCompte();
            consult.setDs(ds);
            consult.setNoDeCompte(noDeCompte);
            consult.consultation();
            req.setAttribute("NoDeCompte",
                            noDeCompte);

            req.setAttribute("Nom",
                            consult.getNom());
            req.setAttribute("Prenom",
                            consult.getPrenom());
            req.setAttribute("Solde",
                            consult.getSolde());
        }
    } catch (...) {
        ...
    }
    getServletContext().getRequestDispatcher(
        "/SaisieNoDeCompte").forward(req, resp);
}

```

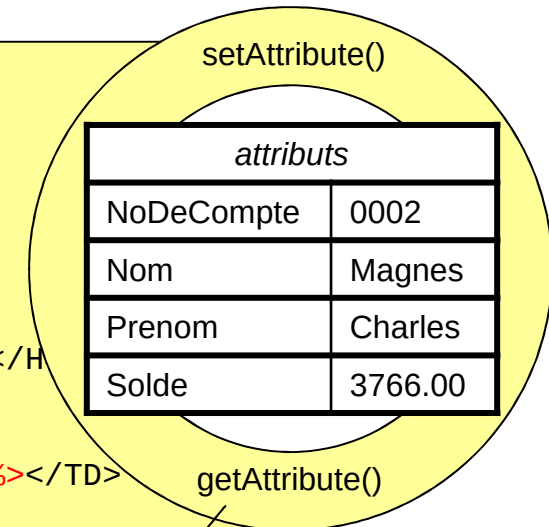


La Servlet (1er scénario)

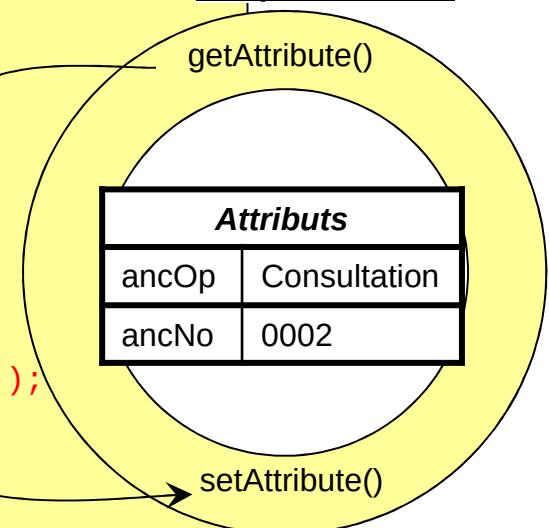
SaisieNoDeCompte.jsp

```
.....
<BODY>
<CENTER>
<FORM action="/Compte/Gestion" method="Post"><BR>
.....
</FORM>
<%if (request.getMethod().equals("POST")) { %>
    <H4><U>Compte N° <%=request.getAttribute("NoDeCompte")%></U></H4>
    <TABLE border="1"><TBODY>
    <TR>
        <TD align="center"><%=request.getAttribute("NoDeCompte")%></TD>
        <TD align="center"><%=request.getAttribute("Nom")%></TD>
        <TD align="center"><%=request.getAttribute("Prenom")%></TD>
        <TD align="center"><%=request.getAttribute("Solde")%></TD>
    </TR>
    </TBODY></TABLE>
<% } %>
<% String ancOp = (String) session.getAttribute("ancOp");
String ancNo = (String) session.getAttribute("ancNo");
if (ancOp != null && ancNo != null){ %>
    <BR><U>Opération précédente: </U><B><%=ancOp%>
    du Compte <%=ancNo%></B>
<% }
session.setAttribute("ancOp", request.getParameter("Demande"));
session.setAttribute("ancNo", request.getAttribute("NoDeCompte"));
%>
</CENTER>
</BODY>
</HTML>
```

HttpServletRequest



HttpSession



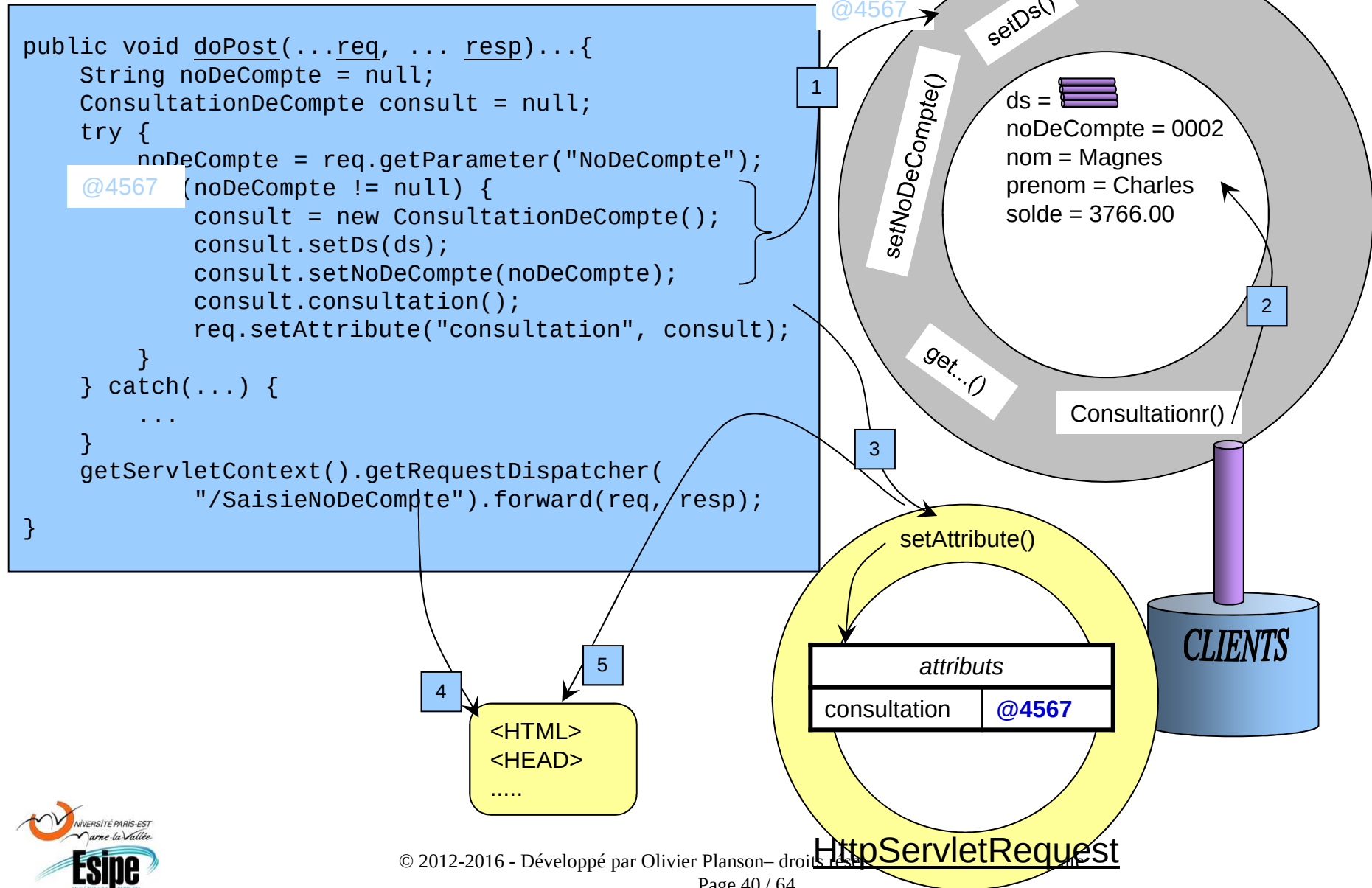
1

2

3

4

La Servlet (2ème scénario....)



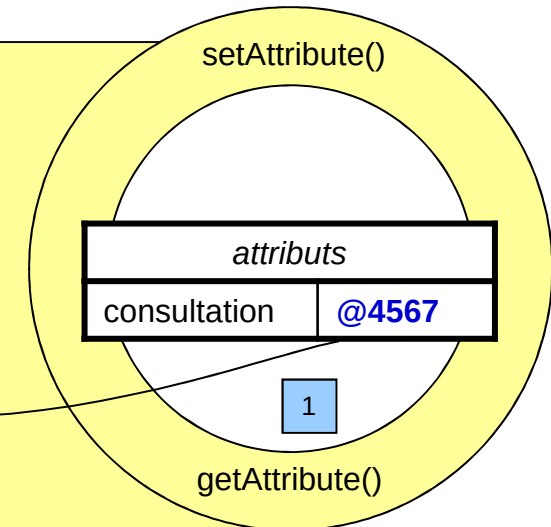


La Servlet (2ème scénario)

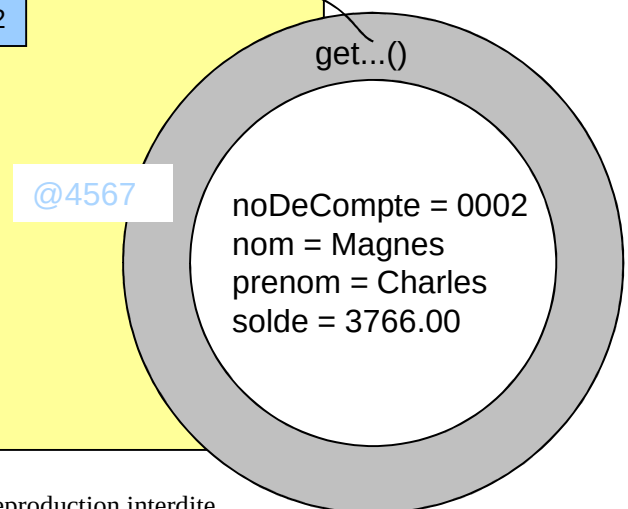
SaisieNoDeCompte.jsp

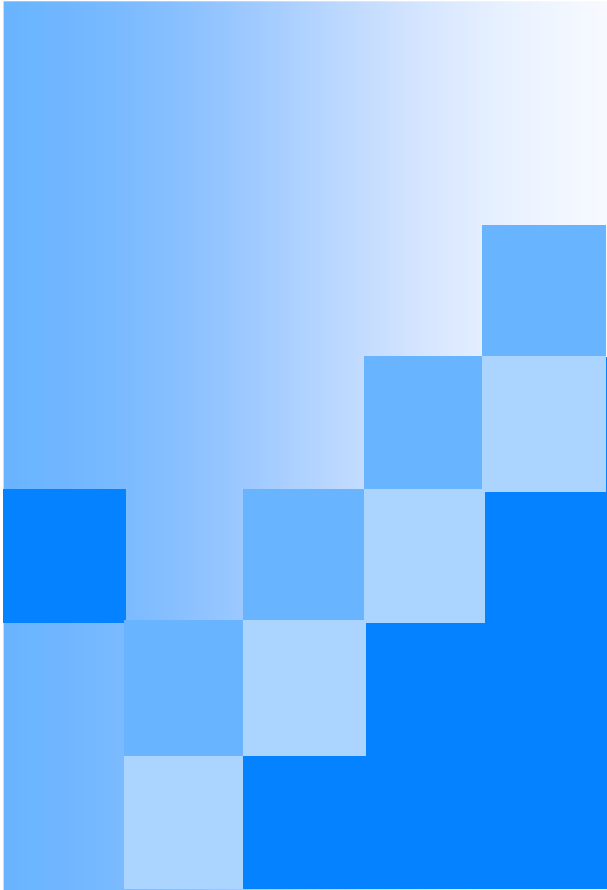
```
.....
<BODY>
<CENTER>
<FORM action="/Compte/Gestion" method="Post"><BR>
.....
</FORM>
<%if (request.getMethod().equals("POST")) { %>
  <jsp:useBean id="consultation" type="Compte.ConsultationDeCompte"
    scope="request"></jsp:useBean>
  <H4><U>Compte N° <%=consultation.getNoDeCompte()%></U></H4>
  <TABLE border="1"><TBODY>
  <TR>
    <TD align="center"><%=consultation.getNoDeCompte()%></TD>
    <TD align="center"><%=consultation.getNom()%></TD>
    <TD align="center"><%=consultation.getPrenom()%></TD>
    <TD align="center"><%=consultation.getSolde()%></TD>
  </TR>
</TBODY></TABLE>
<% } %>
<% String ancOp = (String) session.getAttribute("ancOp");
String ancNo = (String) session.getAttribute("ancNo");
if (ancOp != null && ancNo != null){ %>
  <BR><U>Opératon précédente: </U><B><%=ancOp%>
  du Compte <%=ancNo%></B>
<% }
  session.setAttribute("ancOp", request.getParameter("Demande"));
  session.setAttribute("ancNo", request.getParameter("NoDeCompte"));
%>
.....
```

HttpServletRequest



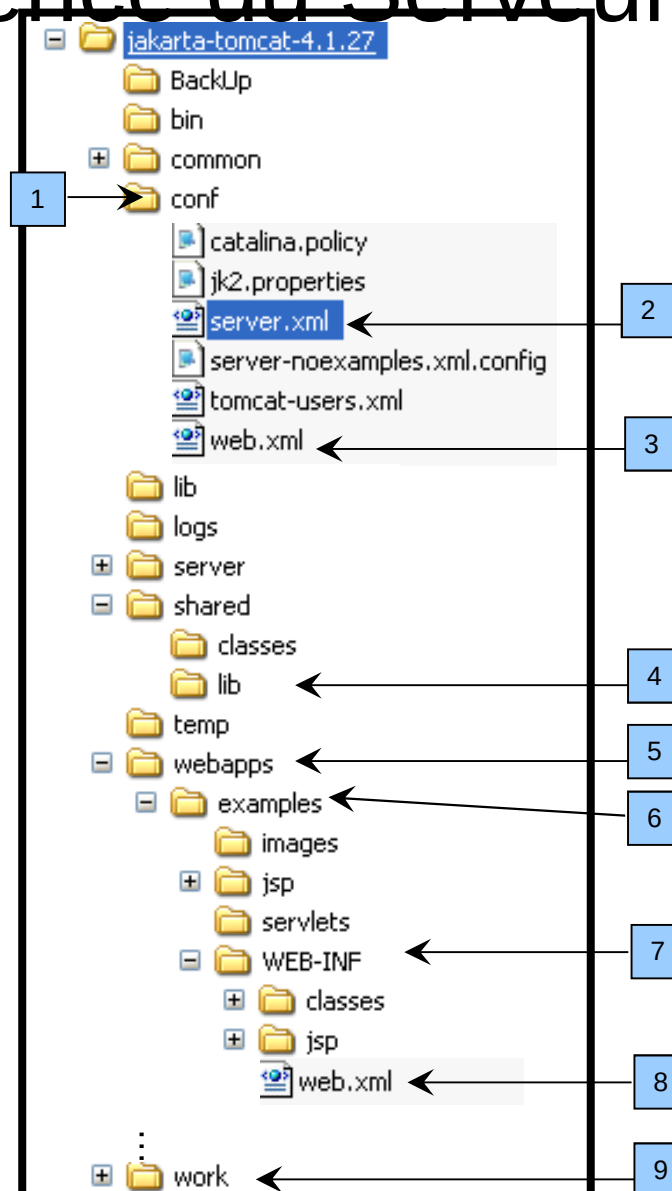
ConsultationDeCompte





Paramétrage d'une Web Application dans un Serveur

L'Arborescence du Serveur





Le Fichier "serveur.xml" de Tomcat

```
<Server port="8005" shutdown="SHUTDOWN" debug="0"> 1
  <Listener className="org.apache.catalina.mbeans.ServerLifecycleListener"
    debug="0"/> 2
  .....
  <Service name="Tomcat-Standalone">
    <Connector className="org.apache.coyote.tomcat4.CoyoteConnector" 3
      port="8080" ..../>
    <Engine name="Standalone" defaultHost="localhost" debug="0"> 4
      ....
      <Host name="localhost" debug="0" appBase="webapps" 5
        unpackWARs="true" autoDeploy="true">
          ...
          <Context path="/examples" docBase="examples" debug="0" 6
            reloadable="true" crossContext="true">
              ...
            </Context>
          </Host>
        </Engine>
      </Service>
    </Server>
```



Le Fichier « web.xml »

```
<?xml version="1.0" encoding="ISO-8859-1"?> 1

<!DOCTYPE web-app
    PUBLIC "-//Sun Microsystems, Inc.//DTD Web Application 2.3//EN"
    "http://java.sun.com/dtd/web-app_2_3.dtd">

<web-app> 2
...
    <filter>
        ...
    </filter>

    <servlet> 4
        ...
    </servlet>
    <servlet-mapping>
        ...
    </servlet-mapping>
    <mime-mapping>
        ...
    </mime-mapping>

</web-app> 3
```



Les Elements du Fichier «web.xml» (1)





Les Eléments du Fichier «web.xml» (2)

```
<session-config>
  <session-timeout>30</session-timeout>
</session-config>

<welcome-file-list>
  <welcome-file>index.html</welcome-file>
  <welcome-file>index.htm</welcome-file>
  <welcome-file>index.jsp</welcome-file>
</welcome-file-list>

<mime-mapping>
  <extension>abs</extension>
  <mime-type>audio/x-mpeg</mime-type>
</mime-mapping>
<mime-mapping>
  <extension>ai</extension>
  <mime-type>application/postscript</mime-type>
</mime-mapping>
<error-page>
  <error-code>401</error-code>
  <location>/NonAutorise.html</location>
</error-page>
<error-page>
  <exception-type>exceptions.TraitementException</exception-type>
  <location>/PbDeTraitement.jsp</location>
</error-page>
```

10

11

12

13

14



Extraits du fichier « web.xml »

1

```
<servlet>
  <servlet-name>default</servlet-name>
  <servlet-class>org.apache.catalina.servlets.DefaultServlet</servlet-class>
...
</servlet>
<servlet>
  <servlet-name>invoker</servlet-name>
  <servlet-class>org.apache.catalina.servlets.InvokerServlet</servlet-class>
...
</servlet>
<servlet>
  <servlet-name>jsp</servlet-name>
  <servlet-class>org.apache.jasper.servlet.JspServlet</servlet-class>
...
</servlet>
<servlet-mapping>
  <servlet-name>default</servlet-name>
  <url-pattern>/</url-pattern>
</servlet-mapping>
<servlet-mapping>
  <servlet-name>invoker</servlet-name>
  <url-pattern>/servlet/*</url-pattern>
</servlet-mapping>
<servlet-mapping>
  <servlet-name>jsp</servlet-name>
  <url-pattern>*.jsp</url-pattern>
</servlet-mapping>
```

2

3

4

5

6

Localisation d'un servlet (TOMCAT)

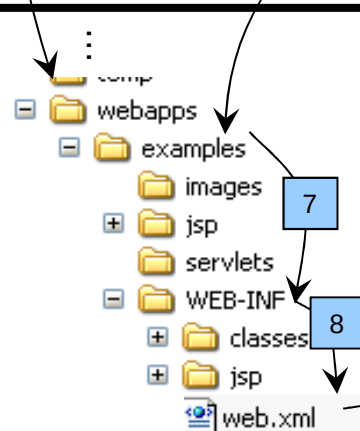
1 <http://localhost:8080/examples/snoop>

server.xml

```
.....
<Service name="Tomcat-Standalone">
  <Connector className="org.apache.coyote.tomcat4.CoyoteConnector"
    port="8080" ..../>
  <Engine name="Standalone" defaultHost="localhost" debug="0">
    ....
    <Host name="localhost" debug="0" appBase="webapps" ...>
      ...
      <Context path="/examples" docBase="examples" ... >
        ...
```

web.xml

```
<web-app>
...
<servlet>
  <servlet-name>snoop</servlet-name>
  <servlet-class>SnoopServlet</servlet-class>
</servlet>
...
<servlet-mapping>
  <servlet-name>snoop</servlet-name>
  <url-pattern>/snoop</url-pattern>
</servlet-mapping>
...
```



Localisation d'un servlet anonyme

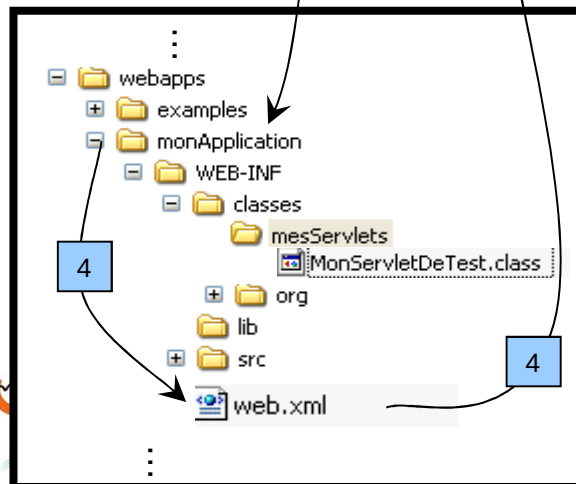
1 <http://localhost:8080/MonAppli/servlet/mesServlets.MonServletDeTest>

2 5 server.xml

```
.....  
<Host name="localhost" debug="0" appBase="webapps" ...>  
  ...  
    <Context path="/examples" docBase="examples" ... >  
    <Context path="/MonAppli" docBase="monApplication" ...>  
  ...
```

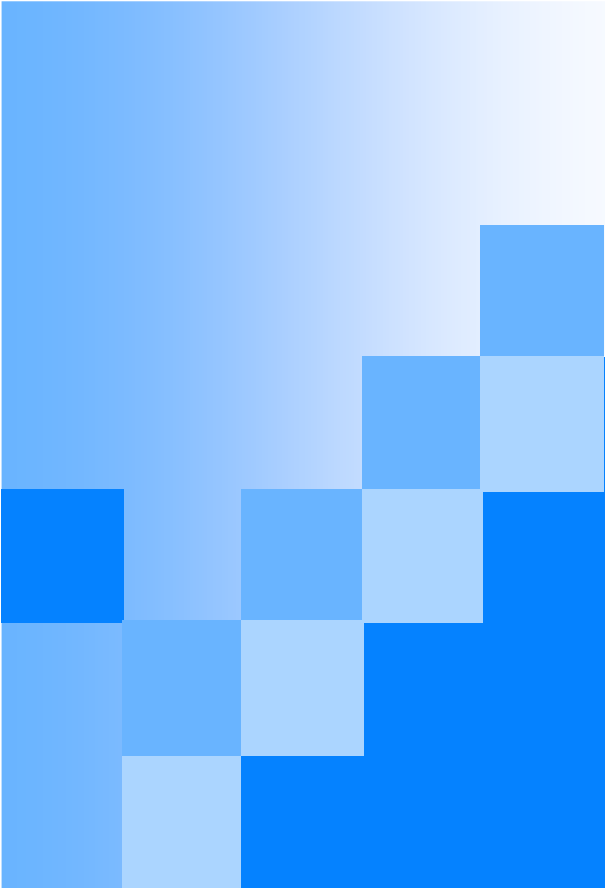
3 web.xml de l'application

```
<web-app>  
  ...  
  <servlet-mapping>  
    <servlet-name>invoker</servlet-name>  
    <url-pattern>/servlet/*</url-pattern>  
  </servlet-mapping>  
  ...
```



6 7 8 web.xml du serveur

```
...  
<servlet>  
  <servlet-name>invoker</servlet-name>  
  <servlet-class>  
    org.apache.catalina.servlets.InvokerServlet  
  </servlet-class>  
  ...  
</servlet>
```



Les balises JSPs



Les balises standards

Les commentaires:

- `<!-- commentaires à faire apparaître dans la page HTML... -->`
- `<!-- commentaires contenant un <%= contenu variable %> ... -->`
- `<%-- commentaire JSP ... --%>`
- `<% /** même effet qu'un commentaire JSP ... **/ %>`

Les déclarations:

- `<%! int i = 0; %>`
`<%! public String f(int i) { if (i<3) return("..."); ... } %>`

Les expressions:

- `<%= expression java %>`

Les scriptlets:

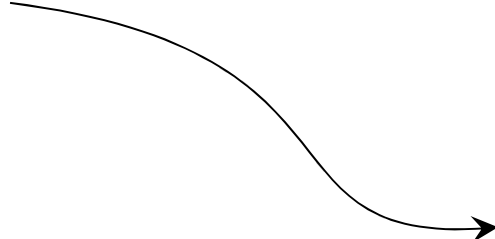
- `<%`
code Java.....
- `%>`



Les directives

Les directives de page:

- `<%@ page page_directive %>`



```
language=" scriptingLanguage"  
extends=" className"  
import=" importList"  
session="true|false"  
buffer="none| sizekb"  
autoFlush="true| false"  
isThreadSafe="true|false"  
info=" info_text"  
errorPage=" error_url"  
isErrorPage="true|false"  
contentType="ctinfo"
```

La directive d'inclusion:

- `<%@ include file="copyright.html" %>`



Autres balises

L'accès aux JavaBeans:

- `<jsp:useBean id=" name" scope=`

```
"page" class="className" ou/et    />  
"requesttype="typeName" ou/et  
"sessionbeanName="beanName"  
"application"
```

L'accès aux propriétés du JavaBean:

- `<jsp:setProperty name="monbean" property="*" />`
 `<jsp:setProperty name="user" property="user" param="username" />`
 `<jsp:setProperty name="resultats" property="rang" value="<%= i+1 %>" />`
- `<jsp:getProperty name="user" property="name" />`

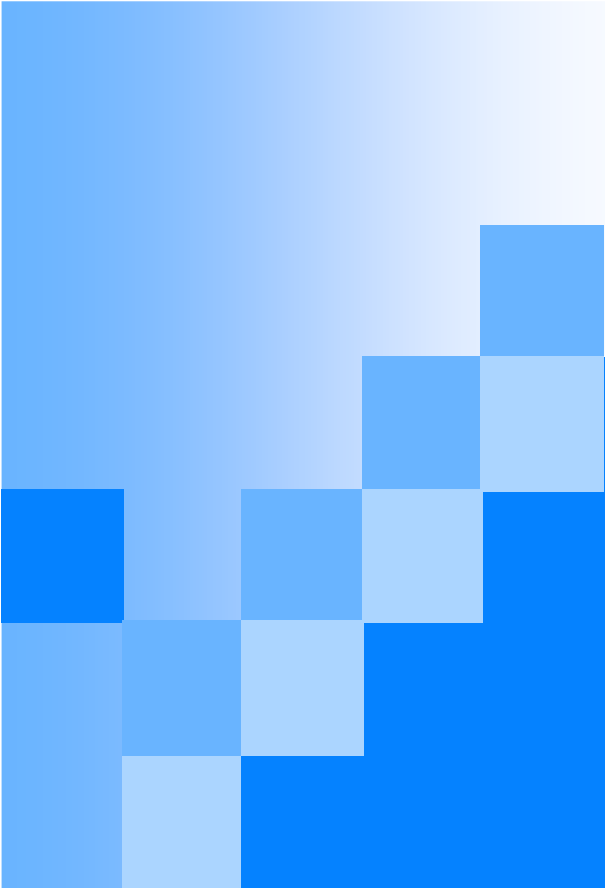
L'appel d'autres pages:

- `<jsp:include page="/templates/copyright.html"/>`
- `<jsp:forward page="/jsps/autreJSP.jsp"/>`
- `<jsp:forward page=" /jsps/autreJSP.jsp"><jsp:param name="addparam" value="valeur"/>`
 `</jsp:forward>`



Les objets implicites

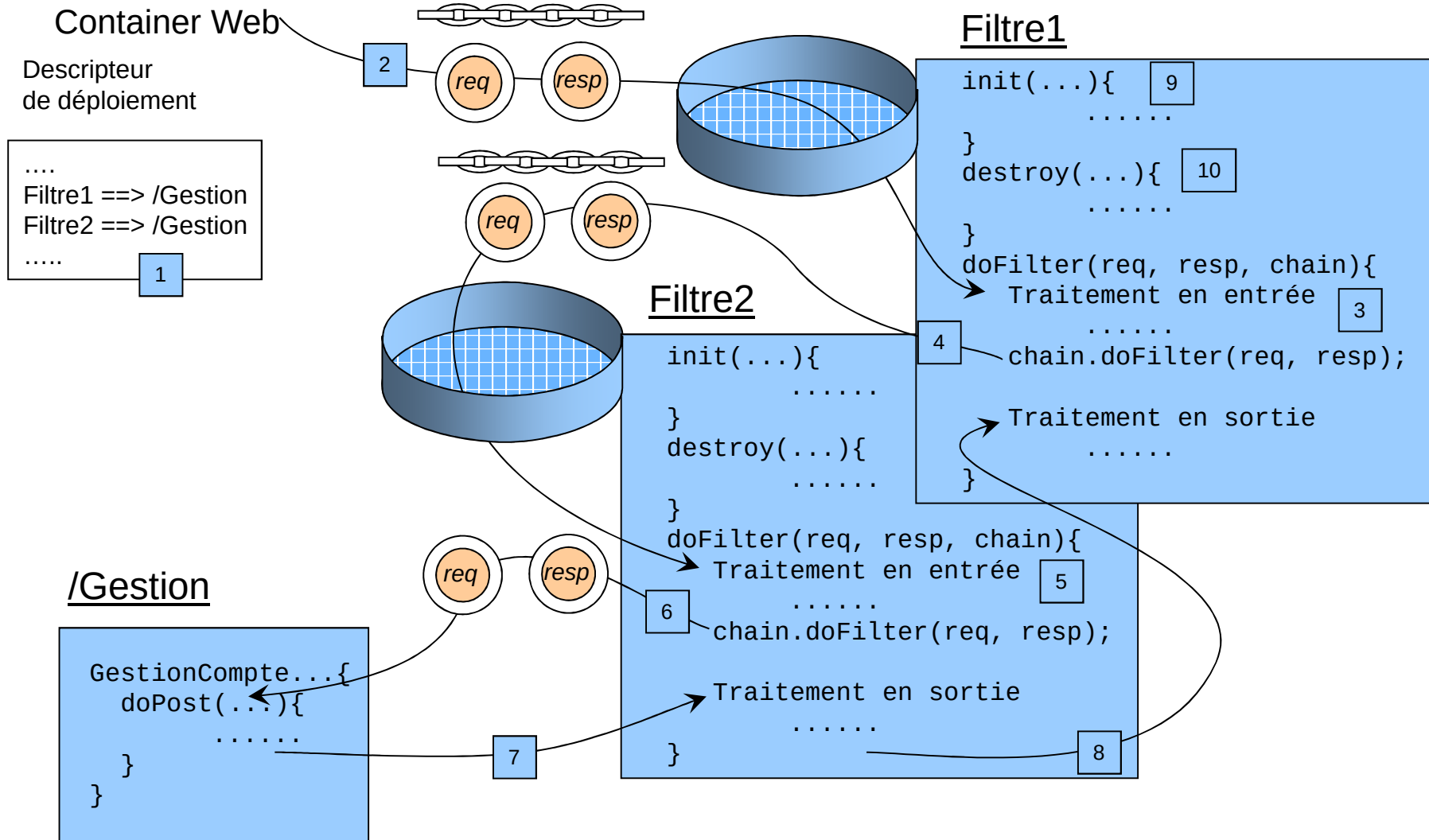
Nom	Type	Durée
request	javax.servlet.HttpServletRequest	request
response	javax.servlet.HttpServletResponse	page
pageContext	javax.servlet.jsp.PageContext	page
session	javax.servlet.http.HttpSession	session
application	javax.servlet.ServletContext	application
out	javax.servlet.jsp.JspWriter	page
config	javax.servlet.ServletConfig	page
page	java.lang.Object	page
exception	java.lang.Throwable	page



Les Filtres

"Filter"

Les Filtres





Extrait d'exemple (1)

ExampleFilter.java

```
public final class ExampleFilter implements Filter {

    private FilterConfig filterConfig = null;

    public void init(FilterConfig filterConfig) throws ServletException {
        this.filterConfig = filterConfig;
    }

    public void destroy() {
        this.filterConfig = null;
    }

    public void doFilter(ServletRequest request, ServletResponse response, FilterChain chain)
        throws IOException, ServletException {

        if (filterConfig == null) return;
        long startTime = System.currentTimeMillis();
        chain.doFilter(request, response);

        long stopTime = System.currentTimeMillis();
        filterConfig.getServletContext().log(this.toString() + ": " + (stopTime - startTime) +
                                                "milliseconds");
    }
}
```





Extrait d'exemples (2)

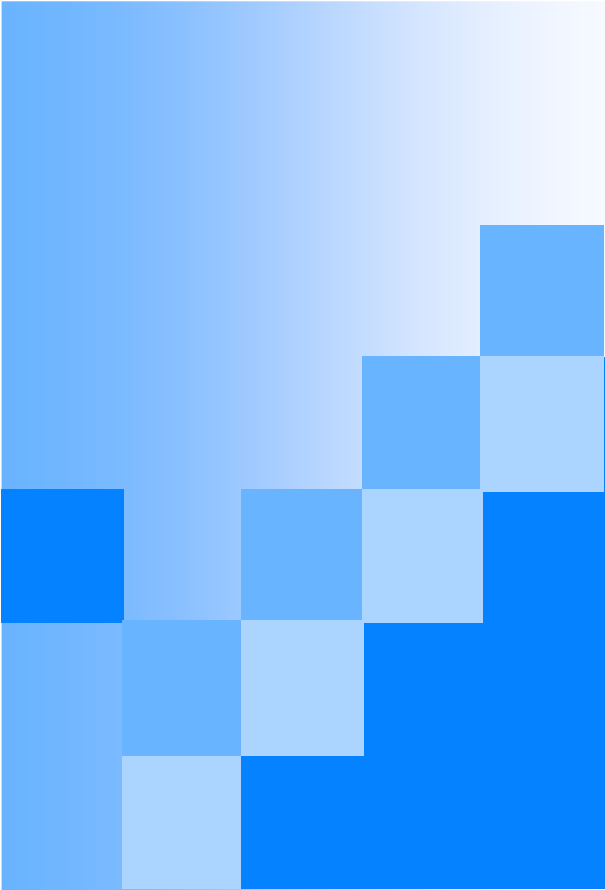
SetCharacterEncodingFilter.java

```
public final class SetCharacterEncodingFilter implements Filter {  
  
    private FilterConfig filterConfig = null;  
    private String encoding = null;  
  
    public void init(FilterConfig filterConfig) throws ServletException {  
        this.filterConfig = filterConfig;  
        this.encoding = filterConfig.getInitParameter("encoding");  
    }  
  
    public void destroy() {  
        this.encoding = null;  
        this.filterConfig = null;  
    }  
  
    public void doFilter(ServletRequest request, ServletResponse response, FilterChain chain)  
        throws IOException, ServletException {  
        if (request.getCharacterEncoding() == null) {  
            request.setCharacterEncoding(this.encoding);  
        }  
    }  
}
```



Le fichier Web.xml

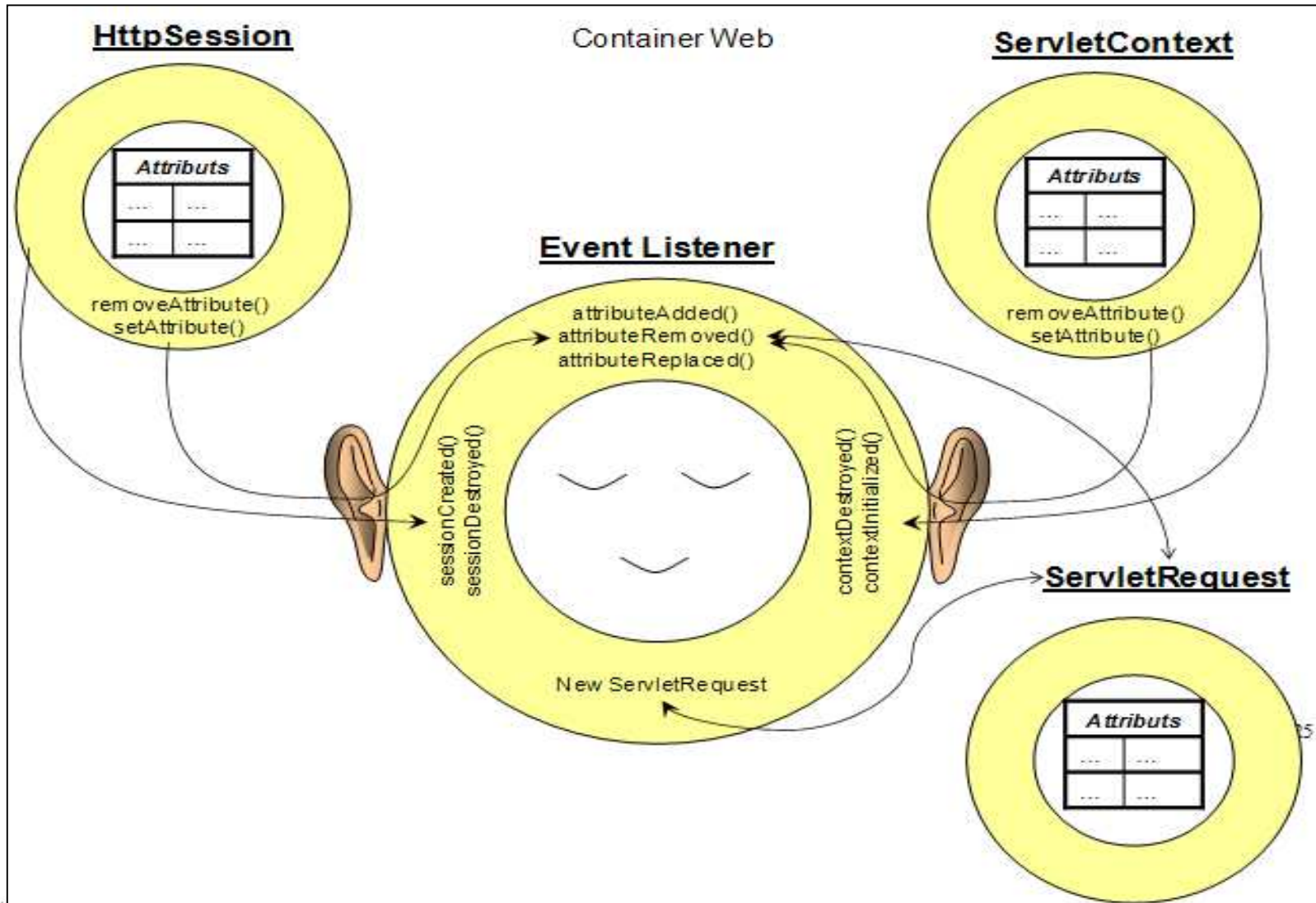
```
<filter>
  <filter-name>Filtre Exemple</filter-name>
  <filter-class>filters.ExampleFilter</filter-class>
</filter>
<filter>
  <filter-name>Filtre Character Encoding</filter-name>
  <filter-class>filters.SetCharacterEncodingFilter</filter-class>
  <init-param>
    <param-name>encoding</param-name>
    <param-value>ISO-8859-1</param-value>
  </init-param>
</filter>
<filter-mapping>
  <filter-name>Filtre Character Encoding</filter-name>
  <url-pattern>/Servlet1</url-pattern>
</filter-mapping>
<filter-mapping>
  <filter-name>Filtre Character Encoding</filter-name>
  <url-pattern>/Servlet2</url-pattern>
</filter-mapping>
<filter-mapping>
  <filter-name> Filtre Exemple</filter-name>
  <url-pattern>/Servlet2</url-pattern>
</filter-mapping>
```



Les Ecouteurs d'Evènements

"Listener"

Les « Listeners »





La Structure d'un « Listener »

```
public class TestListener implements
    ServletContextListener,
    ServletContextAttributeListener,
    HttpSessionListener,
    HttpSessionAttributeListener {

    public void sessionDestroyed(HttpSessionEvent arg0) {
        arg0.getSession();
        ...
    }
    public void sessionCreated(HttpSessionEvent arg0) {
        arg0.getSession();
        ...
    }
    public void attributeRemoved(HttpSessionBindingEvent arg0) {
        arg0.getSession();
        arg0.getName();
        arg0.getValue();
        ...
    }
    public void attributeAdded(HttpSessionBindingEvent arg0) {
        arg0.getSession();
        arg0.getName();
        arg0.getValue();
        ...
    }
    public void attributeReplaced(HttpSessionBindingEvent arg0) {
        arg0.getSession();
        arg0.getName();
        arg0.getValue();
        ...
    }
    ...
}
```



Le fichier Web.xml

```
<web-app>

    ...
    <listener>
        <listener-class>TestListener</listenerclass>
    </listener>
    <listener>
        <listener-class>MyLoggingModule</listener-class>
    </listener>
    ...
</web-app>
```